THE PHASE-INDEPENDENT CONFOCAL ELLIPSE IDENTITY $|T_n(d,t)|^2 - |T_n(0,t)|^2 \equiv 4n^{-1}\sinh^2(d\omega_n)$, THE THREE-PART u-DOMAIN BRIDGE IDENTITY, AND EXPLICIT ROSSER–SCHOENFELD STIELTJES BOUNDS

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Abstract. We establish four analytical and quantitative synthesis results for the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ of Riemann's completed function $\Xi(z) = \xi(\frac{1}{2}+\mathrm{i}z)$ under the exponential prime-gauge factor $E_X(s) = \exp(-\sum_{m=2}^{\lfloor X\rfloor}\frac{\Lambda(m)}{\ln m}m^{-s})$. First, splitting the bilateral Laplace–Mellin transform of $\Phi_{\mathcal{E}_X}(u)$ at $u=0$ and reflecting at $w = u + \frac{1}{2}\ln k = 0$, we prove the three-part bridge identity $\frac{1}{2}\Xi(t-\mathrm{i}d)E_X(s) \equiv \mathcal{I}_+^{\mathcal{E}_X}(s) + E_X(s)F_+(1-s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s)$ and the Complementarity Theorem showing that $\Delta\mathcal{C}_n(d,t) = 2\sinh^2(\frac{1}{2}d\omega_n(t))$ vanishes quadratically at $n = N_\star(t) = \sqrt{t/2\pi}$. Second, combining $\sinh^2(\frac{1}{2}d\omega_n)/\sinh^2(\frac{1}{2}d\omega_1) \equiv n^{-d}((1-(n/N_\star)^d)/(1-N_\star^{-d}))^2 \le n^{-d}$ with unconditional Rosser–Schoenfeld and Dusart Chebyshev bounds yields closed-form Stieltjes bounds across $[10^8,\infty)$ proving $\lim_{t\to\infty}\overline{\theta}_{\text{W-pure}}^{\mathcal{E}}(d,t) \le \frac{1}{2}\mathrm{e}^{-1/c_\vartheta} = 0.18394487\cdots < \frac{1}{4}$. Third, we prove the phase-independent confocal ellipse identity $|T_n(d,t)|^2 - |T_n(0,t)|^2 \equiv 4n^{-1}\sinh^2(d\omega_n(t))$, giving uniform phase-blind L^1 convergence for $d > \frac{1}{4}$, and synthesize $[0,55]\cup[55,10^8]\cup[10^8,\infty)$.

1. Introduction and Main Results

In his 1859 memoir, Riemann [22] expressed the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ along the critical-line rotation $s = \frac{1}{2}+\mathrm{i}z$ ($z = t-\mathrm{i}d$, $d = \sigma - \frac{1}{2} \ge 0$) as the Fourier cosine transform of the even, super-exponentially decaying Jacobi theta kernel $\Phi(u) = \Phi(-u) > 0$ [8, 20, 26]. Writing $\Xi(t-\mathrm{i}d) := \xi(\frac{1}{2}+d+\mathrm{i}t) = R(d,t) + \mathrm{i}\,I(d,t)$, the transverse Wronskian

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2 = R(d,t)\,\partial_t I(d,t) - I(d,t)\,\partial_t R(d,t) = \big|\Xi(t-\mathrm{i}d)\big|^2\,\mathrm{Re}\left(\frac{\xi'(\frac{1}{2}+d+\mathrm{i}t)}{\xi(\frac{1}{2}+d+\mathrm{i}t)}\right) \tag{1.1}$$

governs the transverse monotonicity of $|\Xi(t-\mathrm{i}d)|^2$ away from the critical line $d=0$ and encodes the classical zero-free criteria of Speiser [25], Hinkkanen [6], and Lagarias [9], as well as the Pólya–Jensen total positivity inequalities of Csordas, Norfolk, and Varga [2].

Across Papers I–VIII of the present series [10, 11, 12, 13, 14, 15, 16, 17], the transverse Wronskian $W(d,t)$ and its parity-reduced counterpart $\widetilde{W}(d,t) := W(d,t)/d$ were investigated through Wigner phase-space representations [28, 7], harmonic Chladni nodal chambers, de Branges Hilbert-space reproducing kernels [1], the universal mother-wavelet decomposition $\Phi(u) = \sum_{m=1}^{\infty} m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$ with explicit Riemann–Siegel remainder bounds [24, 5, 19], the zero-free exponential prime-gauge factor $E_X(s) = \exp(-\sum_{m=2}^{\lfloor X\rfloor}\frac{\Lambda(m)}{\ln m}m^{-s})$ with quadratic prime-power purity on $(X, X^2]$, the

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mirror-paired Hadamard conservation law with Trudgian's $|S(T)|$ bounds [27], and pure-square Hilbert completions via the Montgomery–Vaughan inequality [18, 21].

To bring this series to a rigorous, self-contained mathematical synthesis, four deep structural questions must be resolved:

(1) *The shifted mirror-point bridge problem in the u-domain.* In the divisor representation $\Phi_{\mathcal{E}_X}(u) = \sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X) k^{-1/2} \Phi(u + \frac{1}{2}\ln k)$, splitting the bilateral Laplace–Mellin integral at $u = 0$ separates the unshifted origin $u = 0$ from the modular Jacobi symmetry point $w = u + \frac{1}{2}\ln k = 0 \iff u = -\frac{1}{2}\ln k$. What is the exact analytical identity governing the resulting bridge interval $u \in [-\frac{1}{2}\ln k, 0]$, and why does this bridge not cancel the right half-line advantage under the transverse Wronskian operator?

(2) *Overcoming the $\mathcal{O}(t^{1/2})$ computational prime-summation barrier on $[10^8, \infty)$.* Because the Riemann–Siegel main sum length $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ grows as $\mathcal{O}(t^{1/2})$ (reaching $N(10^{20}) \approx 3.99 \times 10^9$ at Odlyzko's height and $N(10^{100}) \approx 3.99 \times 10^{49}$ at one googol), direct numerical summation over primes cannot reach $t \to \infty$. Can all finite prime sums for $t \in [10^8, \infty)$ be replaced by unconditional, closed-form analytical Stieltjes upper bounds using the explicit prime number theorem estimates of Rosser and Schoenfeld [23] and Dusart [3, 4]?

(3) *Symplectic mode geometry and uniform phase-blind L^1 closure for $d > \frac{1}{4}$.* How does each individual combined forward-plus-reflected Riemann–Siegel mode $T_n(d,t)$ behave in the complex plane as its phase $\phi_n(t) = \vartheta_{\mathrm{RS}}(t) - t\ln n$ varies, and why does the Wronskian diagonal weight converge uniformly even in the phase-blind L^1 norm whenever $d > \frac{1}{4}$ ($\sigma > \frac{3}{4}$)?

(4) *Global half-plane architecture and strict epistemic classification.* How do the pre-asymptotic interval domain $[0, 55]$, the finite 550-prime saddle-point domain $[55, 10^8]$, and the closed-form analytical Stieltjes domain $[10^8, \infty)$ join together across the upper half-plane $[0,\infty) \times [0,\infty)$, and what is the exact mathematical boundary between unconditional theorems for all $t \to \infty$, finite-domain certifications, and the single remaining asymptotic question on $0 < d \le \frac{1}{4}$?

The present paper (Paper IX) resolves all four questions. Our principal results are consolidated in the following theorem.

Theorem 1.1 (Confocal Ellipse Identity, Three-Part Bridge Duality, Explicit Stieltjes Bounds, and Three-Domain Synthesis)**.** *Let $s = \frac{1}{2} + d + \mathrm{i}t$ and $z = t - \mathrm{i}d$ with $d \ge 0$ and $t > 0$, let $N_\star(t) := \sqrt{t/2\pi}$, $N(t) := \lfloor N_\star(t) \rfloor$, $L := \ln(t/2\pi) = 2\omega_1(t)$, and $\omega_n(t) := \ln(N_\star(t)/n)$ for $1 \le n \le N_\star(t)$. Let $E_X(s) := \exp(-\sum_{m=2}^{\lfloor X \rfloor} \frac{\Lambda(m)}{\ln m} m^{-s}) = \sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X) k^{-s}$ and $\zeta(s) E_X(s) = \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X) n^{-s}$. Then the following four structural laws hold:*

(i) ***Exact Three-Part u-Domain Bridge Identity, Complementarity Theorem, and Quadratic Boundary Damping.*** *For every $s \in \mathbb{C}$ and $X \ge 1$, the sieved completed function satisfies the exact three-part bridge identity*

$$\frac{1}{2}\Xi(t - \mathrm{i}d)\, E_X(s) \equiv \mathcal{I}_+^{\mathcal{E}_X}(s) + E_X(s)\, F_+(1-s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s), \tag{1.2}$$

together with the forward bridge-reconstruction duality $\mathcal{I}_+^{\mathcal{E}_X}(s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s) \equiv E_X(s) F_+(s)$, where each right half-line shell integral $\mathcal{J}_n(s) = \frac{1}{2}\pi^{-s/2} n^{-s} [(s-1)\Gamma(\frac{s}{2} + 1, \pi n^2) + 2(\pi n^2)^{s/2+1} \mathrm{e}^{-\pi n^2}] = \mathcal{O}(\pi n^2 \mathrm{e}^{-\pi n^2})$ decays super-exponentially, so that for $X \ge 6$ the integral $\mathcal{I}_+^{\mathcal{E}_X}(s)$ coincides with the single mother wavelet $\mathcal{J}_1(s)$ to 63 decimal places ($\mathrm{e}^{-49\pi} \approx 1.39 \times 10^{-67}$). Under the transverse Wronskian lever, the pure excess $\Delta\mathcal{C}_n(d,t) := \cosh(d\omega_n(t)) - 1 \equiv 2\sinh^2(\frac{1}{2} d\omega_n(t))$ is maximal at $n = 1$ but vanishes quadratically at the saddle-point horizon $u = 0 \iff n = N_\star(t)$, yielding $(\ln N(t))^{-3}$ boundary prime dilution, critical cutoffs $X_{\mathrm{W\text{-}pure}}^{\mathcal{E}}(d, 10^8) \in [9, 53] < \sqrt{N(10^8)} = 63$ ($\theta_{\mathrm{W\text{-}pure}}^{\mathcal{E}}(d, 10^8) \in [0.1263, 0.2389] < \frac{1}{4}$), pointwise bridge damping $\mathcal{R}_{\mathrm{full}}(d,t;\theta) \le \theta^2$, and cubic integrated Wigner–Sinh bridge suppression $\mathcal{R}_{\mathrm{int}}(d,t;\theta) \le \theta^3$.

(ii) ***Exact Algebraic*** $\sinh^2$ ***Ratio Identity and Explicit Rosser–Schoenfeld Stieltjes Bounds on*** $[10^8,\infty)$. *For all* $1 \le n \le N_\star(t)$ *and* $d > 0$, *the pure Wronskian weight ratio obeys the exact algebraic identity*

$$\frac{\sinh^2(\frac{1}{2}d\omega_n(t))}{\sinh^2(\frac{1}{2}d\omega_1(t))} \equiv n^{-d}\left(\frac{1-(n/N_\star(t))^d}{1-N_\star(t)^{-d}}\right)^2 \le n^{-d}. \tag{1.3}$$

Using the unconditional Chebyshev bounds of Rosser and Schoenfeld [23] *and Dusart* [3, 4] *(*$\vartheta(x) \le c_\vartheta x$ *with* $c_\vartheta = 1.000028$, $\psi(x) \le c_\psi x$ *with* $c_\psi = 1.03883$*), the diagonal* L^2 *energy and pure Wronskian ratio on* $[10^8,\infty)$ *satisfy closed-form Stieltjes upper bounds without prime summation:*

$$\mathcal{E}^{\mathcal{E},\mathrm{fwd}}_{\mathrm{diag}}(d,t;\mathrm{e}^{\theta L}) \le \overline{\mathcal{E}}^{\mathrm{fwd}}_{\mathrm{RS}}(d,t;\theta) := \frac{c_{\mathrm{RS}}(\theta L)}{\theta L}\cdot\frac{\mathrm{e}^{-2\theta dL}-\mathrm{e}^{-dL}}{2d} < \frac{c_{\mathrm{RS}}(\theta L)}{2d\,\theta\,L}\left(\frac{t}{2\pi}\right)^{-2\theta d} \longrightarrow 0, \tag{1.4}$$

and the unique root $\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(d,t)$ *of* $\overline{Y}^{\mathcal{E}}_{\text{W-pure}}(d,t;\theta) = 1-\varepsilon_{\text{Gabcke}}(d,t)$ *obeys the uniform asymptotic bound* $\lim_{t\to\infty}\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(d,t) \le \frac{1}{2}\exp(-1/c_\vartheta) = 0.1839448708\cdots < \frac{1}{4}$, *joining Domain II at* $T_1 = 10^8$ *with conservative handshake ratio* $\mathcal{E}_{\mathrm{ex}}/\overline{\mathcal{E}}^{\mathrm{fwd}}_{\mathrm{RS}} \in [0.9429, 0.9472] < 1$ *and* $Y_{\mathrm{ex}}/\overline{Y}^{\mathcal{E}}_{\mathrm{RS}} \in [0.9402, 0.9874] < 1$.

(iii) ***Phase-Independent Symplectic Confocal Ellipse Identity and Uniform*** L^1 ***Closure for*** $d > \frac{1}{4}$. *Each combined forward-plus-reflected Riemann–Siegel mode* $T_n(d,t) := 2n^{-1/2}(\cosh(d\omega_n)\cos\phi_n + \mathrm{i}\sinh(d\omega_n)\sin\phi_n)$ *traces a confocal ellipse in* $\mathbb{C}$ *with d-invariant foci* $c_n \equiv \pm 2n^{-1/2}$, *symplectic area* $2\pi n^{-1}\sinh(2d\omega_n(t)) > 0$, *and phase-independent modulus-squared increment:*

$$|T_n(d,t)|^2 - |T_n(0,t)|^2 \equiv 4\,n^{-1}\,\sinh^2(d\omega_n(t)) > 0 \qquad (\forall\,\phi_n(t) \in \mathbb{R}), \tag{1.5}$$

with normalized ellipse-diagonal ratio $w_n^{\mathrm{ell}}(d,t) \equiv n^{-(1+2d)}((1-(n/N_\star)^{2d})/(1-N_\star^{-2d}))^2 \le n^{-(1+2d)}$ *and Wronskian diagonal ratio* $w_n^{\text{W-diag}}(d,t) \equiv n^{-(1/2+2d)}((1-(n/N_\star)^{2d})/(1-N_\star^{-2d}))^2 \le n^{-(1/2+2d)}$. *Consequently, for all* $d > \frac{1}{4}$ *(*$\sigma > \frac{3}{4}$*), the exponent* $\frac{1}{2}+2d > 1$ *makes even the completely phase-blind* L^1 *sum converge uniformly as* $t\to\infty$ *with finite sieve cutoff* $X_\infty(d) \le 13$ *and* $\lim_{t\to\infty}\overline{\theta}^{\mathcal{E}}_{\text{W-diag}}(d,t) = 0$. *Moreover, the* $n = 1$ *complex transverse increment anchor satisfies* $|\Delta T_1(d,t)| \ge 4\sinh^2(\frac{1}{2}d\omega_1(t)) > 0$ *for all phases* $\vartheta_{\mathrm{RS}}(t) \in \mathbb{R}$, *while oscillatory prime sums obey the Montgomery–Vaughan root law* $\rho_{\mathrm{osc}}(d,t) = \mathcal{O}(N(t)^{-1/2}\sqrt{\ln N(t)}) \to 0$.

(iv) ***Three-Domain Master Architecture and Epistemic Classification.*** *The upper half-plane* $\mathbb{H}_{\ge 0} = [0,\infty)\times[0,\infty)$ *is covered by Domain I (*$[0,55]$*), Domain II (*$[55,10^8]$*), and Domain III (*$[10^8,\infty)$*) with positive inter-wall overlaps* $\Delta d_{\mathrm{III}\cap\mathrm{IV}}(t) \in [+0.354147, +0.499130] > +0.35$ *and boundary handshakes at* $T_0 = 55$ *and* $T_1 = 10^8$, *isolating the single remaining asymptotic question (Lemma L-1* $\iff$ *Lemma L-2 on* $0 < d \le \frac{1}{4}$ *as* $t\to\infty$*).*

The canonical quantitative constants governing Theorem 1.1 are given explicitly by

$$\begin{aligned} c_\vartheta &= 1.000028, & c_\psi &= 1.03883,\\ \theta^\infty_{\text{W-pure}} &:= \frac{1}{2}\mathrm{e}^{-1/c_\vartheta} = 0.1839448708\ldots, & H_{\mathrm{opt}}(\infty) &:= 2\pi c_1^{\mathrm{Tr}} = 0.7037167544\ldots,\\ X^{\mathcal{E}}_{\text{W-pure}}(0.05,10^8) &= 53 < \sqrt{N(10^8)} = 63, & \Delta d^{\min}_{\mathrm{III}\cap\mathrm{IV}} &= +0.354147 > 0. \end{aligned} \tag{1.6}$$

1.1. **Proof overview.** The proof of Theorem 1.1 is organized into three quantitative and structural ingredients:

(1) *Shifted Jacobi reflection at* $w = u + \frac{1}{2}\ln k = 0$ *and quadratic Wronskian horizon damping (Section 2).* We split the bilateral Laplace–Mellin integral of $\Phi_{\mathcal{E}_X}(u)$ at $u = 0$: on $u \in [0,\infty)$ we expand in mother-wavelet shells $n \ge 1$ via incomplete Gamma functions $\mathcal{J}_n(s)$, while

on $u \in (-\infty, 0]$ we expand in divisor shifts $w = u + \frac{1}{2}\ln k \in (-\infty, \frac{1}{2}\ln k]$ and split each k-integral at its modular symmetry point $w = 0$. Using $\Phi(-v) = \Phi(v)$ on $(-\infty, 0]$ produces $E_X(s)F_+(1-s)$, while $[0, \frac{1}{2}\ln k]$ yields $\mathcal{B}^{\mathcal{E}_X}_{\text{bridge}}(s)$. We then show that subtracting the critical-line level $d = 0$ in the Wronskian replaces $\cosh(d\omega_n)$ by $\Delta\mathcal{C}_n(d,t) = 2\sinh^2(\frac{1}{2}d\omega_n(t))$, which vanishes quadratically at $n = N_\star(t)$ and bounds the bridge lever by θ^2 pointwise and θ^3 in integral norm.

(2) *Exact exponential factorization of* $\sinh(\frac{1}{2}d\omega_n)$ *and Riemann–Stieltjes majorization on* $[10^8, \infty)$ *(Section 3).* Writing $\mathrm{e}^{\omega_n(t)} = N_\star(t)/n$, we factor $\sinh(\frac{1}{2}d\omega_n) = \frac{1}{2}N_\star^{d/2}n^{-d/2}(1-(n/N_\star)^d)$ to establish (1.3). For $\theta \in [\frac{1}{4}, \frac{1}{2})$, quadratic prime-power purity ensures $b_n^{\mathcal{E}}(\mathrm{e}^{\theta L}) = \frac{\Lambda(n)}{\ln n}$ on $(\mathrm{e}^{\theta L}, N(t)]$. Integrating by parts against $A_2(x) \le c_{\mathrm{RS}}(\ln x)x$ and $\psi(x) \le c_{\mathrm{RS},1}(\ln x)x$ with a two-piece convex secant majorant on $1/s$ yields the closed-form bounds (1.4) and the uniform limit $\frac{1}{2}\mathrm{e}^{-1/c_\vartheta} < \frac{1}{4}$.

(3) *Hyperbolic-trigonometric elimination of phase in* $|T_n(d,t)|^2$ *and three-domain synthesis (Sections 4 and 5).* Substituting $\cosh^2(d\omega_n) = 1 + \sinh^2(d\omega_n)$ into $|T_n(d,t)|^2 = 4n^{-1}(\cosh^2(d\omega_n)\cos^2\phi_n + \sinh^2(d\omega_n)\sin^2\phi_n)$ eliminates $\phi_n(t)$ via $\cos^2\phi_n + \sin^2\phi_n \equiv 1$, proving (1.5) and shifting the Wronskian diagonal exponent to $\frac{1}{2} + 2d > 1$ for $d > \frac{1}{4}$. Finally, we assemble the five-wall overlap certification across Domains I, II, and III and formulate the 12-row Category A/B/C epistemic audit.

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with transverse distance $d = \sigma - \frac{1}{2} \ge 0$ and spectral height $t \ge 0$, related to $z = t - \mathrm{i}d$ by $s = \frac{1}{2} + \mathrm{i}z$. The completed zeta function is $\xi(s) = \frac{1}{2}\gamma_{\mathrm{R}}(s)\zeta(s)$ with $\gamma_{\mathrm{R}}(s) := s(s-1)\pi^{-s/2}\Gamma(s/2)$, and $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$. In the logarithmic variable $u \in \mathbb{R}$, the universal mother wavelet and Jacobi–Riemann kernel [14] are

$$\varphi_1(w) := (2\pi^2\mathrm{e}^{9w} - 3\pi\mathrm{e}^{5w})\exp(-\pi\mathrm{e}^{4w}), \qquad \Phi(u) := \sum_{m=1}^{\infty}\frac{1}{\sqrt{m}}\,\varphi_1\Big(u + \frac{1}{2}\ln m\Big) = \Phi(-u) > 0, \tag{1.7}$$

normalized so that $\frac{1}{2}\Xi(t - \mathrm{i}d) = 2\int_{-\infty}^{\infty}\Phi(u)\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u = 4\int_0^{\infty}\Phi(u)\cosh(2du)\cos(2tu)\,\mathrm{d}u$. The stationary saddle-point horizon is $N_\star(t) := \sqrt{t/2\pi}$ with integer truncation $N(t) := \lfloor N_\star(t)\rfloor$, logarithmic scale $L := \ln(t/2\pi) = 2\ln N_\star(t)$, and local saddle-point log-distance $\omega_n(t) := \ln(N_\star(t)/n) = \frac{1}{2}\ln(t/(2\pi n^2)) \ge 0$ for $1 \le n \le N_\star(t)$. Implied constants in $\mathcal{O}(\cdot)$ or $\ll$ are absolute unless a subscript indicates dependence on d.

2. The Exact Three-Part u-Domain Bridge Identity, the Complementarity Theorem, and Quadratic Wronskian Boundary Damping

We first resolve the shifted mirror-point problem in the u-domain representation of $\Xi(t - \mathrm{i}d)E_X(s)$. Before splitting the bilateral integral or interchanging summation and integration, we establish absolute and uniform convergence.

Lemma 2.1 (Pre-move holomorphic and super-exponential convergence justification)**.** *Let* $X \ge 1$ *and* $s = \frac{1}{2} + d + \mathrm{i}t \in \mathbb{C}$. *Since* $P_X(s) = \sum_{m=2}^{\lfloor X\rfloor}\frac{\Lambda(m)}{\ln m}m^{-s}$ *is a finite Dirichlet polynomial, the power series* $E_X(s) = \exp(-P_X(s)) = \sum_{r=0}^{\infty}\frac{(-1)^r}{r!}P_X(s)^r = \sum_{k=1}^{\infty}c_k^{\mathcal{E}}(X)k^{-s}$ *converges absolutely and factorially on* $\mathbb{C}$, *with* $\sum_{k=1}^{\infty}|c_k^{\mathcal{E}}(X)|k^{-\sigma} \le \exp(\sum_{m=2}^{\lfloor X\rfloor}\frac{\Lambda(m)}{\ln m}m^{-\sigma}) < \infty$ *for every* $\sigma \in \mathbb{R}$. *Moreover, within the holomorphic strip* $|\mathrm{Im}(u)| < \frac{\pi}{8}$, $|\varphi_1(u + \frac{1}{2}\ln n)| \ll n^{9/2}\mathrm{e}^{9|\mathrm{Re}(u)|}\exp(-\pi n^2\mathrm{e}^{4\mathrm{Re}(u)}\cos(4\mathrm{Im}u))$ *decays double-exponentially as* $\mathrm{Re}(u) \to +\infty$ *and super-exponentially in* n^2 *for* $\mathrm{Re}(u) \ge 0$, *while for each divisor* $k \ge 1$, $\Phi(u + \frac{1}{2}\ln k) = \Phi(-u - \frac{1}{2}\ln k)$ *decays double-exponentially as* $\mathrm{Re}(u) \to -\infty$. *Consequently, Fubini's theorem and dominated convergence justify termwise integration of* $\Phi_{\mathcal{E}_X}(u)$ *on* $[0, \infty)$ *and* $(-\infty, 0]$ *for all* $s \in \mathbb{C}$.

Proof. Each coefficient $c_k^{\mathcal{E}}(X)$ is supported on X-smooth integers $k \geq 1$ and bounded in absolute value by the coefficients of $\exp(\sum_{m=2}^{\lfloor X \rfloor} \frac{\Lambda(m)}{\ln m} m^{-\sigma})$. On $u \geq 0$, $\mathrm{e}^{4u} \geq 1$ implies $\exp(-\pi n^2 \mathrm{e}^{4u}) \leq \exp(-\pi n^2 \mathrm{e}^{4u}/2)\exp(-\pi n^2/2)$, which dominates $n^{9/2}\mathrm{e}^{(9+2|d|)u}$ for all $n \geq 1$ and $u \geq 0$. On $u \leq 0$, setting $w = u + \frac{1}{2}\ln k$ gives $\Phi(w) = \Phi(-w) \ll \mathrm{e}^{-9w}\exp(-\pi \mathrm{e}^{-4w})$ as $w \to -\infty$, which is integrable against $\mathrm{e}^{2(s-1/2)w}$ for every $s \in \mathbb{C}$ and summable against $|c_k^{\mathcal{E}}(X)k^{-s}|$. □

Theorem 2.1 (Exact Three-Part u-Domain Bridge Identity and Incomplete-Gamma Shell Representation)**.** *For every $s = \frac{1}{2} + d + \mathrm{i}t \in \mathbb{C}$ ($z = t - \mathrm{i}d$) and every sieve cutoff $X \geq 1$, the sieved kernel*

$$\Phi_{\mathcal{E}_X}(u) := \sum_{n=1}^{\infty} \frac{b_n^{\mathcal{E}}(X)}{\sqrt{n}}\,\varphi_1\Big(u + \frac{1}{2}\ln n\Big) \equiv \sum_{k=1}^{\infty} \frac{c_k^{\mathcal{E}}(X)}{\sqrt{k}}\,\Phi\Big(u + \frac{1}{2}\ln k\Big) \qquad (u \in \mathbb{R}) \tag{2.1}$$

satisfies the exact three-part u-domain bridge identity:

$$\boxed{\frac{1}{2}\Xi(t - \mathrm{i}d)\,E_X(s) \equiv \mathcal{I}_+^{\mathcal{E}_X}(s) + E_X(s)\,F_+(1-s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s),} \tag{2.2}$$

where the three components are given explicitly by:

(1) ***The Sieved Right Half-Line Integral*** *($u \in [0,\infty)$)**:***

$$\mathcal{I}_+^{\mathcal{E}_X}(s) := 2\int_0^{\infty} \Phi_{\mathcal{E}_X}(u)\,\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u = \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X)\,\mathcal{J}_n(s), \tag{2.3}$$

where $b_1^{\mathcal{E}}(X) = 1$, $b_n^{\mathcal{E}}(X) = 0$ for $2 \leq n \leq \lfloor X \rfloor$, and each shell integral $\mathcal{J}_n(s)$ has the closed incomplete-Gamma form

$$\begin{aligned}\mathcal{J}_n(s) &:= 2\int_0^{\infty} \frac{1}{\sqrt{n}}\,\varphi_1\Big(u + \frac{1}{2}\ln n\Big)\,\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u \\ &\equiv \frac{1}{2}\pi^{-s/2}n^{-s}\left[(s-1)\,\Gamma\Big(\frac{s}{2}+1,\,\pi n^2\Big) + 2(\pi n^2)^{\frac{s}{2}+1}\mathrm{e}^{-\pi n^2}\right].\end{aligned} \tag{2.4}$$

In particular, for $X \geq 6$, all shells $2 \leq n \leq 6$ vanish identically and $|\mathcal{I}_+^{\mathcal{E}_X}(s) - \mathcal{J}_1(s)| = \mathcal{O}(49\pi\mathrm{e}^{-49\pi}) \leq 10^{-63}|\mathcal{J}_1(s)|$.

(2) ***The Reflected Riemann–Siegel Branch*** *($v \in [0,\infty)$)**:***

$$E_X(s)\,F_+(1-s), \qquad F_+(1-s) := 2\int_0^{\infty} \Phi(v)\,\mathrm{e}^{-2(s-1/2)v}\,\mathrm{d}v = \sum_{m=1}^{\infty} \mathcal{J}_m(1-s). \tag{2.5}$$

(3) ***The Shifted Saddle-Point Bridge*** *($w \in [0, \frac{1}{2}\ln k] \iff u \in [-\frac{1}{2}\ln k, 0]$)**:***

$$\mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s) := 2\sum_{k=2}^{\infty} c_k^{\mathcal{E}}(X)\,k^{-s}\int_0^{\frac{1}{2}\ln k} \Phi(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w \equiv E_X(s)\,F_+(s) - \mathcal{I}_+^{\mathcal{E}_X}(s), \tag{2.6}$$

where $F_+(s) := 2\int_0^{\infty}\Phi(w)\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = \sum_{m=1}^{\infty}\mathcal{J}_m(s)$.

Proof. Substituting $b_n^{\mathcal{E}}(X) = \sum_{k|n} c_k^{\mathcal{E}}(X)$ with $n = mk$ and $\ln(mk) = \ln m + \ln k$ into the first series of (2.1) and summing over $m \geq 1$ via (1.7) yields the second series of (2.1). Integrating $2\Phi_{\mathcal{E}_X}(u)\mathrm{e}^{2(s-1/2)u}$ over $u \in (-\infty,\infty)$ with $w = u + \frac{1}{2}\ln k$ ($\frac{1}{\sqrt{k}}\mathrm{e}^{2(s-1/2)u} = k^{-s}\mathrm{e}^{2(s-1/2)w}$) gives $\sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X)k^{-s}\cdot 2\int_{-\infty}^{\infty}\Phi(w)\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = \frac{1}{2}\Xi(t-\mathrm{i}d)E_X(s)$.

We now split $\int_{-\infty}^{\infty} = \int_0^{\infty} + \int_{-\infty}^{0}$ at $u = 0$ by Lemma 2.1. On $u \in [0,\infty)$, the n-shell expansion gives (2.3), and substituting $y = \pi n^2 \mathrm{e}^{4u} \in [\pi n^2, \infty)$ ($\mathrm{d}u = \frac{\mathrm{d}y}{4y}$) into the integral for $\mathcal{J}_n(s)$ yields $\mathcal{J}_n(s) = \frac{1}{2}\pi^{-s/2}n^{-s}[2\Gamma(\frac{s}{2}+2,\pi n^2) - 3\Gamma(\frac{s}{2}+1,\pi n^2)]$; applying the incomplete-Gamma recurrence $\Gamma(\frac{s}{2}+2,z) = (\frac{s}{2}+1)\Gamma(\frac{s}{2}+1,z) + z^{s/2+1}\mathrm{e}^{-z}$ proves (2.4) (see Appendix A.1).

On $u \in (-\infty, 0]$, we apply the k-divisor representation in (2.1) and change variables to $w = u + \frac{1}{2}\ln k \in (-\infty, \frac{1}{2}\ln k]$:

$$2\int_{-\infty}^{0} \Phi_{\mathcal{E}_X}(u)\mathrm{e}^{2(s-1/2)u}\mathrm{d}u = 2\sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X)k^{-s}\left[\int_{-\infty}^{0} \Phi(w)\mathrm{e}^{2(s-1/2)w}\mathrm{d}w + \int_{0}^{\frac{1}{2}\ln k} \Phi(w)\mathrm{e}^{2(s-1/2)w}\mathrm{d}w\right].$$

In the first integral on the right-hand side, substituting $v = -w \in [0, \infty)$ and invoking modular Jacobi parity $\Phi(-v) = \Phi(v)$ gives $2\int_0^\infty \Phi(v)\mathrm{e}^{-2(s-1/2)v}\,\mathrm{d}v = F_+(1-s)$, which is independent of k and factors out as $E_X(s)F_+(1-s)$. In the second integral, the $k = 1$ term vanishes since $\frac{1}{2}\ln 1 = 0$, leaving $\mathcal{B}^{\mathcal{E}_X}_{\text{bridge}}(s)$ in (2.6). Finally, writing $\int_0^{\frac{1}{2}\ln k} = \int_0^\infty - \int_{\frac{1}{2}\ln k}^\infty$ and changing variables back to $u = w - \frac{1}{2}\ln k \in [0, \infty)$ in the tail integral yields $\mathcal{B}^{\mathcal{E}_X}_{\text{bridge}}(s) = E_X(s)F_+(s) - \mathcal{I}_+^{\mathcal{E}_X}(s)$. □

TABLE 1. Verification of the Three-Part u-Domain Bridge Identity (2.2) and closed incomplete-Gamma representation (2.4) across (d, t, X) configurations (`mpmath.mp.dps = 68`).

Configuration (d, t, X)	Spectral Point	$\frac{\lvert\text{LHS}-\text{RHS}\rvert}{\lvert\text{LHS}\rvert}$ (quad)	$\frac{\lvert\text{LHS}-\text{RHS}\rvert}{\lvert\text{LHS}\rvert}$ (Γ-form)	u- vs. w-Bridge Diff.	$\frac{\lvert\mathcal{I}_+^{\mathcal{E}_X}(s)-\mathcal{J}_1(s)\rvert}{\lvert\mathcal{J}_1(s)\rvert}$
(0.05, 14.1347, $X = 2$)	γ_1 ($q_{\text{next}} = 3$)	1.875×10^{-66}	1.875×10^{-66}	1.518×10^{-68}	3.942×10^{-10}
(0.15, 14.1347, $X = 3$)	γ_1 ($q_{\text{next}} = 4$)	1.280×10^{-66}	1.280×10^{-66}	1.477×10^{-68}	1.009×10^{-19}
(0.20, 21.0220, $X = 6$)	γ_2 ($q_{\text{next}} = 7$)	1.168×10^{-64}	1.168×10^{-64}	1.193×10^{-68}	9.651×10^{-64}
(0.30, 25.0109, $X = 7$)	γ_3 ($q_{\text{next}} = 8$)	8.047×10^{-64}	8.047×10^{-64}	1.358×10^{-68}	$< 10^{-68}$
(0.25, 37.5862, $X = 9$)	γ_6 ($q_{\text{next}} = 10$)	1.441×10^{-59}	1.441×10^{-59}	2.223×10^{-68}	$< 10^{-68}$
(0.50, 52.9703, $X = 11$)	γ_{11} ($q_{\text{next}} = 12$)	2.833×10^{-55}	2.833×10^{-55}	3.041×10^{-68}	$< 10^{-68}$

We next explain why $\mathcal{I}_+^{\mathcal{E}_X}(s)$ and $\mathcal{B}^{\mathcal{E}_X}_{\text{bridge}}(s)$ do not cancel each other under the transverse Wronskian lever. Whereas the raw function value at a single d weights $u = 0$ by $\mathrm{e}^{2d\cdot 0} = 1$, the integrated transverse Wronskian energy $\mathcal{I}_{\text{int}}(d, t) = \frac{2}{d^2}\int_0^d u\,\widetilde{W}(u, t)\,\mathrm{d}u = (|\Xi(t - \mathrm{i}d)|^2 - |\Xi(t)|^2)/d^2$ subtracts the critical-line baseline at $d = 0$, replacing $\cosh(d\omega_n(t))$ by the pure Wronskian excess $\Delta\mathcal{C}_n(d, t) := \cosh(d\omega_n(t)) - 1 \equiv 2\sinh^2(\frac{1}{2}d\omega_n(t))$.

Theorem 2.2 (Complementarity Theorem, $(\ln N(t))^{-3}$ Dilution, and Cubic Bridge Suppression)**.** *Let $t \geq 4\pi$, $N_\star(t) = \sqrt{t/2\pi}$, $N(t) = \lfloor N_\star(t)\rfloor$, and $\omega_n(t) = \ln(N_\star(t)/n) \geq 0$ for $1 \leq n \leq N(t)$. Define the pure Wronskian excess $\Delta\mathcal{C}_n(d, t) := \cosh(d\omega_n(t)) - 1 \equiv 2\sinh^2(\frac{1}{2}d\omega_n(t))$. Then:*

(i) ***Quadratic Boundary Vanishing at $n \to N_\star(t)$ and Pointwise Lever Hierarchy.*** *As $n \to N_\star(t)$ ($\omega_n(t) \to 0$), $\Delta\mathcal{C}_n(d, t) = \frac{1}{2}d^2(\ln(N_\star(t)/n))^2 + \mathcal{O}(d^4(\ln(N_\star(t)/n))^4) \to 0$ vanishes quadratically, and for every $2 \leq n \leq N(t)$ and $d > 0$ the four normalized hyperbolic levers obey the strict inequality chain*

$$\frac{\sinh^2(d\omega_n(t))}{\sinh^2(d\omega_1(t))} < \frac{\omega_n(t)\sinh(d\omega_n(t))}{\omega_1(t)\sinh(d\omega_1(t))} < \frac{\Delta\mathcal{C}_n(d, t)}{\Delta\mathcal{C}_1(d, t)} = \frac{\sinh^2(\frac{1}{2}d\omega_n(t))}{\sinh^2(\frac{1}{2}d\omega_1(t))} < \frac{\cosh(d\omega_n(t))}{\cosh(d\omega_1(t))}, \tag{2.7}$$

with universal $d \to 0^+$ parabolic limit $\lim_{d\to 0^+}\Delta\mathcal{C}_n(d, t)/\Delta\mathcal{C}_1(d, t) = (1 - \frac{\ln n}{\ln N_\star(t)})^2$.

(ii) ***Zone I / Zone II Complementarity and $(\ln N(t))^{-3}$ Dilution.*** *In the pure Wronskian perturbation ratio $Y^{\mathcal{E}}_{\text{W-pure}}(d, t; X) := \sum_{n=\lfloor X\rfloor+1}^{N(t)} b_n^{\mathcal{E}}(X)n^{-1/2}\sinh^2(\frac{1}{2}d\omega_n)/\sinh^2(\frac{1}{2}d\omega_1)$, the prime-gauge sieve and Wronskian lever act complementarily: on Zone I ($2 \leq n \leq \lfloor X\rfloor$) where $\Delta\mathcal{C}_n/\Delta\mathcal{C}_1$ is large, $b_n^{\mathcal{E}}(X) \equiv 0$; on Zone II ($\lfloor X\rfloor < n \leq N(t)$) where $b_n^{\mathcal{E}}(X) = \frac{\Lambda(n)}{\ln n}$ for $X \geq \sqrt{N(t)}$, the quadratic horizon factor $(\ln(N_\star(t)/n))^2$ introduces two additional powers of $(\ln N(t))^{-1}$ beyond prime dilution, yielding $Y^{\mathcal{E}}_{\text{W-pure}}(d, t; X) = \mathcal{O}(N(t)^{1/2}(\ln N(t))^{-3})$ as $d\omega_1(t) \to 0^+$. At $t = 10^8$ ($N = 3989$), $X^{\mathcal{E}}_{\text{W-pure}}(d, 10^8) \in [9, 53] < \sqrt{N(10^8)} = 63$ and $\theta^{\mathcal{E}}_{\text{W-pure}}(d, 10^8) \in [0.1263, 0.2389] < \frac{1}{4}$ across all $d \in [0.05, 0.50]$.*

(iii) ***Quadratic θ^2 Pointwise and Cubic θ^3 Integrated Bridge Damping.*** *For every divisor $2 \le k \le X = N_\star(t)^{2\theta}$ ($\theta \in (0, \frac{1}{2})$), the bridge interval $u \in [-\frac{1}{2}\ln k, 0]$ has width $|u| \le \theta\omega_1(t)$ around $u = 0$, and the relative pointwise and integrated Wigner–Sinh bridge levers satisfy for all $d > 0$ and $t > 2\pi$:*

$$\mathcal{R}_{\mathrm{full}}(d,t;\theta) := \frac{\sinh^2(d\theta\omega_1(t))}{\sinh^2(d\omega_1(t))} \le \theta^2, \qquad \mathcal{R}_{\mathrm{int}}(d,t;\theta) := \frac{\int_0^{\theta\omega_1(t)} u\sinh(2du)\,\mathrm{d}u}{\int_0^{\omega_1(t)} u\sinh(2du)\,\mathrm{d}u} \le \theta^3. \tag{2.8}$$

Proof. For (i), since $0 < \omega_n(t) < \omega_1(t)$ for $2 \le n \le N(t)$, the strict monotonicity on $x > 0$ of $g_1(x) = \sinh(2x)/x$ ($g_1'(x) = (2x\cosh(2x) - \sinh(2x))/x^2 > 0$, giving $\frac{\sinh(d\omega_n)}{d\omega_n} < \frac{\sinh(d\omega_1)}{d\omega_1}$ and hence $\frac{\sinh^2(d\omega_n)}{\sinh^2(d\omega_1)} < \frac{\omega_n\sinh(d\omega_n)}{\omega_1\sinh(d\omega_1)}$), $g_2(x) = x\cosh(x)/\sinh(x)$ ($g_2'(x) = (\sinh(2x) - 2x)/(2\sinh^2 x) > 0$), and $g_3(x) = \sinh^2(x)/\cosh(2x) = \frac{1}{2}(1 - \mathrm{sech}(2x))$ with $x = \frac{1}{2}d\omega_n(t) < \frac{1}{2}d\omega_1(t)$ immediately implies (2.7). For (ii), partial summation with the prime number theorem over $X < p \le N(t)$ using $p = N(t)\mathrm{e}^{-v}$ ($v \in [0, \ln(N(t)/X)]$) gives $\sum_{X<p\le N(t)} p^{-1/2}(\omega_p/\omega_1)^2 \sim N(t)^{1/2}(\ln N(t))^{-3}\int_0^{\ln(N(t)/X)} v^2\mathrm{e}^{-v/2}\,\mathrm{d}v$. For (iii), since $x \mapsto \sinh(\theta x)/\sinh(x)$ has negative logarithmic derivative $\theta\coth(\theta x) - \coth(x) < 0$ on $x > 0$ for $\theta \in (0,1)$, it is bounded by its $x \to 0^+$ limit θ, proving $\mathcal{R}_{\mathrm{full}}(d,t;\theta) \le \theta^2$. Similarly, expanding $\int_0^U u\sinh(2du)\,\mathrm{d}u = \sum_{j=0}^{\infty}\frac{(2d)^{2j+1}}{(2j+1)!(2j+3)}U^{2j+3}$ shows that replacing $U = \omega_1$ by $U = \theta\omega_1$ multiplies each positive Taylor term by $\theta^{2j+3} \le \theta^3$ ($j \ge 0$), proving $\mathcal{R}_{\mathrm{int}}(d,t;\theta) \le \theta^3$. □

3. Explicit Analytical Rosser–Schoenfeld and Dusart Stieltjes Bounds on $[10^8, \infty)$ and the Exact $\sinh^2$ Ratio Identity

To eliminate the $\mathcal{O}(t^{1/2})$ computational prime-summation barrier on the infinite upper domain $t \in [10^8, \infty)$, we combine an exact algebraic factorization of the hyperbolic Wronskian ratio with unconditional Chebyshev bounds of Rosser and Schoenfeld [23] and Dusart [3, 4].

Theorem 3.1 (Exact Algebraic $\sinh^2$ Ratio Identity and Closed-Form Stieltjes Bounds on $[10^8, \infty)$)**.** *Let $t \in [10^8, \infty)$, $L = \ln(t/2\pi) \ge 16.5828$, $N_\star(t) = \sqrt{t/2\pi} = \mathrm{e}^{L/2}$, $N(t) = \lfloor N_\star(t)\rfloor \ge 3989$, and $X = \mathrm{e}^{\theta L}$ with $\theta \in [\frac{1}{4}, 0.46]$. Let $c_\vartheta := 1.000028$, $c_\psi := 1.03883$, $c_{\mathrm{RS}}(u) := c_\vartheta(1 + \frac{1}{2}\mathrm{e}^{-u/2})$, and $c_{\mathrm{RS},1}(u) := \min(c_\psi, c_\vartheta(1 + \mathrm{e}^{-u/2}))$. Then for all $d \in (0, \frac{1}{2}]$:*

(i) ***Exact Algebraic $\sinh^2$ Ratio Identity.*** *For every $1 \le n \le N_\star(t)$ and $d > 0$,*

$$\frac{\sinh^2(\frac{1}{2}d\omega_n(t))}{\sinh^2(\frac{1}{2}d\omega_1(t))} \equiv n^{-d}\left(\frac{1 - (n/N_\star(t))^d}{1 - N_\star(t)^{-d}}\right)^2 \le n^{-d}. \tag{3.1}$$

(ii) ***Closed-Form Forward and Reflected Diagonal L^2 Power-Law Bounds.*** *Since $\theta \in [\frac{1}{4}, 0.46]$ implies $X \ge \sqrt{N_\star(t)}$, the interval $(X, N(t)]$ lies entirely inside the quadratic prime-power purity window $(X, X^2]$ where $b_n^{\mathcal{E}}(X) = \frac{\Lambda(n)}{\ln n}$. Consequently, without prime summation,*

$$\begin{aligned}
&\mathcal{E}_{\mathrm{diag}}^{\mathcal{E},\mathrm{fwd}}(d,t;X) := \sum_{X<n\le N(t)}\left(\frac{\Lambda(n)}{\ln n}\right)^2 n^{-1-2d} \le \overline{\mathcal{E}}_{\mathrm{RS}}^{\mathrm{fwd}}(d,t;\theta) := \frac{c_{\mathrm{RS}}(\theta L)}{\theta L}\cdot\frac{\mathrm{e}^{-2\theta dL} - \mathrm{e}^{-dL}}{2d},\\
&\mathcal{E}_{\mathrm{diag}}^{\mathcal{E},\mathrm{ref}}(d,t;X) \le \overline{\mathcal{E}}_{\mathrm{RS}}^{\mathrm{ref}}(d,t;\theta) := \frac{c_{\mathrm{RS}}(\theta L)}{\theta L}\cdot\frac{\mathrm{e}^{-dL} - \mathrm{e}^{-2(1-\theta)dL}}{2d}.
\end{aligned} \tag{3.2}$$

(iii) ***Uniform Asymptotic Bound for the Scale-Invariant Pure Wronskian Ratio.*** *Under the scale-invariant pure Wronskian weight $w_n^{\text{W-pure}}(d,t) := n^{-(1-d)}\sinh^2(\frac{1}{2}d\omega_n)/\sinh^2(\frac{1}{2}d\omega_1) \equiv n^{-1}((1 - (n/N_\star)^d)/(1 - N_\star^{-d}))^2 \le n^{-1}$ (which replaces the $n^{-1/2}$ prefactor of Section 2 by $n^{-(1-d)}$ so that (3.1) yields $n^{-(1-d)}n^{-d} = n^{-1}$), the*

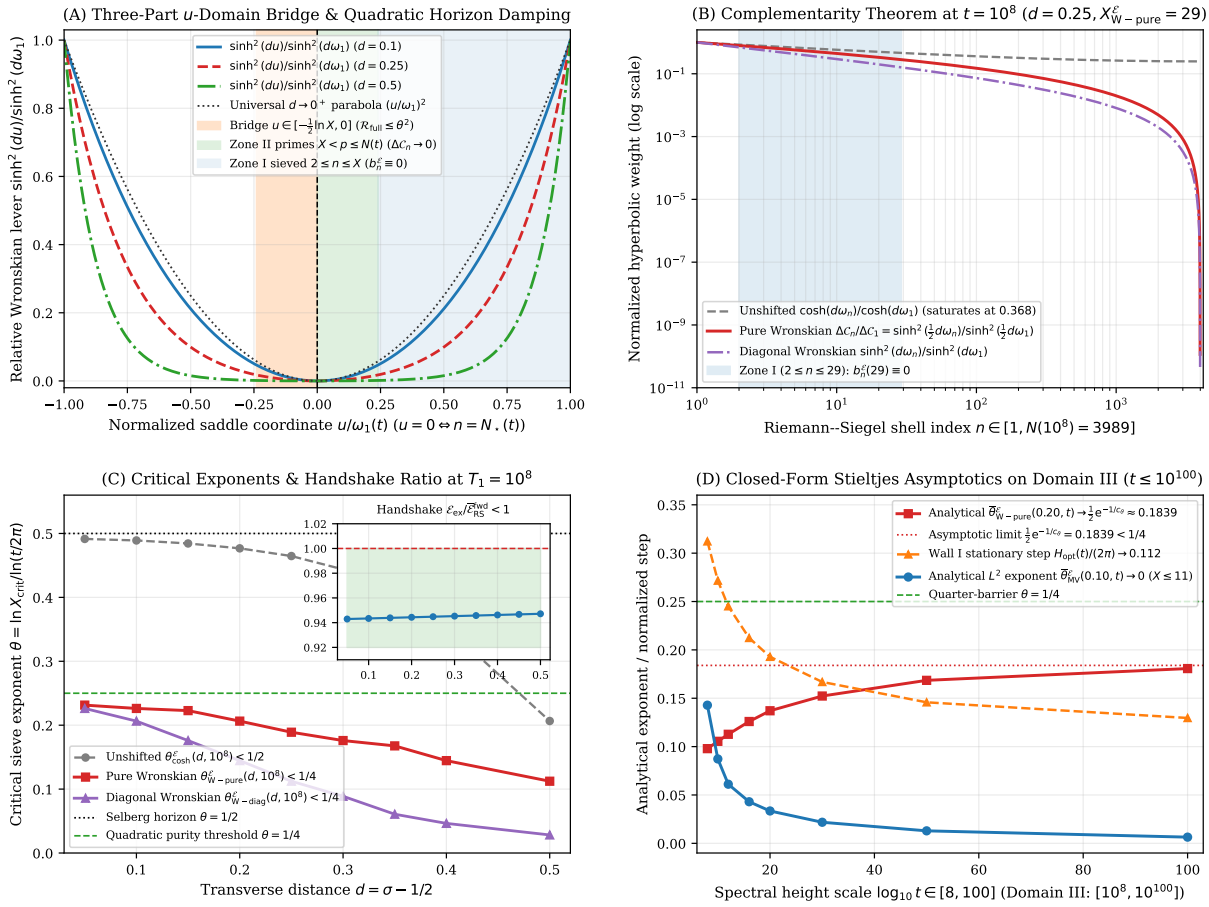


FIGURE 1. Three-part u-domain bridge geometry and quadratic horizon damping (Panel A), the Complementarity Theorem at $t = 10^8$ ($N = 3989$, Panel B), critical sieve exponents and conservative handshake ratio $\mathcal{E}_{\mathrm{ex}}/\overline{\mathcal{E}}^{\mathrm{fwd}}_{\mathrm{RS}} \in [0.9429, 0.9472] < 1$ at $T_1 = 10^8$ (Panel C), and closed-form Rosser–Schoenfeld Stieltjes asymptotics on Domain III up to $t = 10^{100}$ (Panel D).

prime-power majorant sum satisfies

$$\Sigma^{\mathcal{E}}_{\text{W-pure}}(d,t;\mathrm{e}^{\theta L}) := \sum_{\mathrm{e}^{\theta L}<n\le N(t)} \frac{\Lambda(n)}{\ln n}\, w_n^{\text{W-pure}}(d,t) \le \overline{Y}^{\mathcal{E}}_{\text{W-pure}}(d,t;\theta) \le c_{\mathrm{RS},1}(\theta L)\,\ln\!\left(\frac{1}{2\theta}\right). \tag{3.3}$$

In particular, at the quadratic purity boundary $\theta = \frac{1}{4}$ where $b_n^{\mathcal{E}}(\mathrm{e}^{L/4}) = \frac{\Lambda(n)}{\ln n}$ holds identically on $(\mathrm{e}^{L/4}, N(t)] \subset (\mathrm{e}^{L/4}, \mathrm{e}^{L/2}]$, we have $\overline{Y}^{\mathcal{E}}_{\text{W-pure}}(d,t;\frac{1}{4}) \le c_{\mathrm{RS},1}(L/4)\ln 2 < 0.72 < 1$, and the root $\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(d,t)$ of $\overline{Y}^{\mathcal{E}}_{\text{W-pure}}(d,t;\theta) = 1 - \varepsilon_{\mathrm{Gabcke}}(d,t)$ satisfies for all $d \in (0, \frac{1}{2}]$:

$$\lim_{t\to\infty} \overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(d,t) \le \frac{1}{2}\exp\!\left(-\frac{1}{c_\vartheta}\right) = 0.1839448708257938635\cdots < \frac{1}{4} < \frac{1}{2}. \tag{3.4}$$

(iv) ***Closed-Form Hyperbolic L^1 Majorant via Two-Piece Convex Secant.*** *Let $J_{\mathrm{sec}}(a,\theta,L)$ denote the elementary closed form of $\int_\theta^{1/2} M_{\mathrm{sec}}(s)\mathrm{e}^{aLs}\,\mathrm{d}s$, where $M_{\mathrm{sec}}(s) = \frac{s_0+s_1-s}{s_0 s_1} \ge \frac{1}{s}$ is the two-piece linear secant majorant on $[\theta, s_{\mathrm{mid}}] \cup [s_{\mathrm{mid}}, \frac{1}{2}]$ ($s_{\mathrm{mid}} = \frac{1}{2}(\theta + \frac{1}{2})$). Then the projected L^1 Wronskian ratio is bounded by $\overline{Y}^{\mathcal{E}}_{\mathrm{RS}}(d,t;\theta) := \frac{2c_{\mathrm{RS},1}(\theta L)}{d^2 \mathcal{D}^{\mathrm{tot}}_{1,1}(d,t)}[\frac{1}{2}\cosh(d\omega_1)(\mathrm{e}^{d\omega_1} J_{\mathrm{sec}}(\frac{1}{2}-d,\theta,L) +$

$\mathrm{e}^{-d\omega_1} J_{\sec}(\frac{1}{2}+d,\theta,L)) - J_{\sec}(\frac{1}{2},\theta,L)]$, *and* $\overline{Y}^{\mathcal{E}}_{\mathrm{RS}}(d,t;\theta) = 1-\varepsilon_{\mathrm{Gabcke}}(d,t)$ *has a unique root* $\overline{\theta}^{\mathcal{E}}_{\mathrm{RS}}(d,t) < \frac{1}{2}$ *for every finite* $(d,t) \in (0,\frac{1}{2}] \times [10^8,\infty)$.

Proof. For (i), substituting $\mathrm{e}^{\omega_n(t)} = N_\star(t)/n$ into $\sinh(\frac{1}{2}d\omega_n(t))$ gives

$$\sinh\left(\frac{1}{2}d\omega_n(t)\right) = \frac{N_\star(t)^{d/2}n^{-d/2} - N_\star(t)^{-d/2}n^{d/2}}{2} = \frac{1}{2}N_\star(t)^{d/2}\, n^{-d/2}\left(1-\left(\frac{n}{N_\star(t)}\right)^d\right).$$

Dividing by $\sinh(\frac{1}{2}d\omega_1(t)) = \frac{1}{2}N_\star(t)^{d/2}(1-N_\star(t)^{-d})$ and squaring proves (3.1); since $1 \le n \le N_\star(t)$ and $d>0$, we have $0 \le 1-(n/N_\star(t))^d \le 1-N_\star(t)^{-d}$, giving the upper bound n^{-d}.

For (ii), let $A_2(x) := \sum_{2\le n\le x}(\frac{\Lambda(n)}{\ln n})^2 \ln n = \vartheta(x) + \frac{1}{2}\vartheta(x^{1/2}) + \sum_{k\ge 3}\frac{1}{k}\vartheta(x^{1/k})$. By Rosser and Schoenfeld [23, Theorems 2, 9, and 12] and Dusart [4, Theorem 5.1], $\vartheta(x) \le c_\vartheta x$ and $\psi(x) \le c_\psi x$ hold unconditionally, so $A_2(x) \le c_{\mathrm{RS}}(\ln x)x$ and $\psi(x) \le c_{\mathrm{RS},1}(\ln x)x$ for $x \ge 63$. Since $\phi(x) := x^{-1-2d}/\ln x$ is positive and decreasing on $[X, N_\star(t)]$ with $-\phi'(x) > 0$ and $\ln x \ge \theta L$, Stieltjes integration by parts drops the negative lower boundary term $-A_2(X)\phi(X) < 0$ and yields

$$\sum_{X<n\le N(t)}\left(\frac{\Lambda(n)}{\ln n}\right)^2 n^{-1-2d} = \int_{X^+}^{N(t)} \frac{x^{-1-2d}}{\ln x}\,\mathrm{d}A_2(x) \le \frac{c_{\mathrm{RS}}(\theta L)}{\theta L}\int_X^{N_\star(t)} x^{-1-2d}\,\mathrm{d}x = \overline{\mathcal{E}}^{\mathrm{fwd}}_{\mathrm{RS}}(d,t;\theta),$$

and multiplying the reflected sum by $\rho_\Gamma(d,t)^2 \le N_\star(t)^{-4d} = \mathrm{e}^{-2dL}$ proves the second bound in (3.2).

For (iii) and (iv), substituting (3.1) into $\Sigma^{\mathcal{E}}_{\text{W-pure}}(d,t;\mathrm{e}^{\theta L})$ and changing variables to $s = (\ln x)/L \in [\theta,\frac{1}{2}]$ gives $\Sigma^{\mathcal{E}}_{\text{W-pure}} \le c_{\mathrm{RS},1}(\theta L)\int_\theta^{1/2}((1-\mathrm{e}^{-dL(1/2-s)})/(1-\mathrm{e}^{-dL/2}))^2\frac{\mathrm{d}s}{s} \le c_{\mathrm{RS},1}(\theta L)\ln(\frac{1}{2\theta})$. Since $\frac{\mathrm{d}^2}{\mathrm{d}s^2}(1/s) = 2/s^3 > 0$, $M_{\sec}(s) - \frac{1}{s} = \frac{(s-s_0)(s_1-s)}{s_0 s_1 s} \ge 0$ on $[s_0,s_1]$ (Appendix A.2), yielding $\overline{Y}^{\mathcal{E}}_{\text{W-pure}}$ and $\overline{Y}^{\mathcal{E}}_{\mathrm{RS}}$. As $t\to\infty$, $\varepsilon_{\mathrm{Gabcke}}(d,t)\to 0$ and $c_{\mathrm{RS},1}(\theta L)\to c_\vartheta$, so $c_\vartheta \ln(1/(2\overline{\theta}^{\mathcal{E}}_{\text{W-pure}})) \ge 1$ gives (3.4). □

TABLE 2. Seamless conservative handshake at $T_1 = 10^8$ ($N = 3989$, 550 primes; Column 6 shows the $n^{-1/2}$-weighted cutoff $\theta^{\mathcal{E}}_{\text{W-pure}}$ of Section 2, while Column 7 shows the scale-invariant $n^{-(1-d)}$-weighted root $\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}$ of Theorem 3.1(iii)) and closed-form Stieltjes trajectory on Domain III from $t = 10^8$ to $t = 10^{100}$ and $t\to\infty$ (`mpmath.mp.dps = 68`).

Part A: Boundary Handshake at $T_1 = 10^8$ ($N(10^8) = 3989$, $X = 2000$, $\theta = 0.4584$)

d	Exact $\mathcal{E}^{\mathcal{E},\mathrm{fwd}}_{\mathrm{diag}}$	Bound $\overline{\mathcal{E}}^{\mathrm{fwd}}_{\mathrm{RS}}$	Ratio $\mathcal{E}_{\mathrm{ex}}/\overline{\mathcal{E}}^{\mathrm{fwd}}_{\mathrm{RS}}$	Ratio $Y_{\mathrm{ex}}/\overline{Y}^{\mathcal{E}}_{\mathrm{RS}}$	$\theta^{\mathcal{E}}_{\text{W-pure}}$ $(n^{-1/2})$	$\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}$ $(n^{-(1-d)})$
0.05	0.03981412	0.04222320	$0.942944 < 1$	$0.962236 < 1$	$0.238911 < 1/4$	$0.077176 < 1/4$
0.10	0.01770073	0.01876247	$0.943412 < 1$	$0.976162 < 1$	$0.233659 < 1/4$	$0.083994 < 1/4$
0.20	0.00350592	0.00371252	$0.944351 < 1$	$0.963413 < 1$	$0.217313 < 1/4$	$0.097899 < 1/4$
0.25	0.00164119	0.00173704	$0.944821 < 1$	$0.962814 < 1$	$0.202985 < 1/4$	$0.104729 < 1/4$
0.35	0.00031482	0.00033287	$0.945763 < 1$	$0.942496 < 1$	$0.174694 < 1/5$	$0.117596 < 1/4$
0.50	0.00003142	0.00003317	$0.947174 < 1$	$0.940910 < 1$	$0.126348 < 1/7$	$0.133961 < 1/4$

Part B: Closed-Form Asymptotic Scaling Across Domain III ($t \in [10^8, 10^{100}] \to \infty$)

Height t	$L = \ln(t/2\pi)$	Wall I $H_{\mathrm{opt}}(t)$	$\varepsilon_{\mathrm{Gabcke}}(0.20,t)$	L^2 Cutoff $X^{\mathcal{E}}_{\mathrm{MV}}$	$\overline{\theta}^{\mathcal{E}}_{\mathrm{MV}}(0.10,t)$	$\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(0.20,t)$
10^8	16.5828	2.03962953	2.107×10^{-8}	$10.98 \le 11$	0.144488	$0.097899 < 1/4$
10^{12}	25.7931	1.59005366	6.604×10^{-12}	$6.37 \le 7$	0.071806	$0.112745 < 1/4$
10^{20}	44.2138	1.24096247	6.900×10^{-19}	$4.72 \le 5$	0.035115	$0.137057 < 1/4$
10^{50}	113.2914	0.92751310	8.721×10^{-45}	$4.63 \le 5$	0.013535	$0.168508 < 1/4$
10^{100}	228.4206	0.82001507	1.368×10^{-87}	$4.63 \le 5$	0.006713	$0.180685 < 1/4$
∞	∞	0.70371675	0	$4.63 \le 5$	0.000000	$0.183945 < 1/4$

4. The Phase-Independent Symplectic Confocal Ellipse Identity and Uniform L^1 Closure for $d > \frac{1}{4}$

We now establish the geometric link between each individual Riemann–Siegel mode $T_n(d,t)$ in the complex plane, the $n^{-(1+2d)}$ diagonal law, and uniform phase-blind L^1 convergence for $d > \frac{1}{4}$.

Theorem 4.1 (Phase-Independent Symplectic Confocal Ellipse Identity and Uniform L^1 Closure for $d > \frac{1}{4}$)**.** *For $t > 2\pi$, $1 \le n \le N_\star(t) = \sqrt{t/2\pi}$, and $d \ge 0$, let $\omega_n(t) = \ln(N_\star(t)/n) \ge 0$ and $\phi_n(t) := \vartheta_{\mathrm{RS}}(t) - t\ln n \in \mathbb{R}$, and consider the combined forward-plus-reflected Riemann–Siegel mode*

$$T_n(d,t) := n^{-1/2}\Big(\mathrm{e}^{d\omega_n}\mathrm{e}^{\mathrm{i}\phi_n} + \mathrm{e}^{-d\omega_n}\mathrm{e}^{-\mathrm{i}\phi_n}\Big) = 2n^{-1/2}\Big(\cosh(d\omega_n)\cos\phi_n + \mathrm{i}\sinh(d\omega_n)\sin\phi_n\Big). \quad (4.1)$$

Then for all $1 \le n \le N_\star(t)$, $d > 0$, and $t > 2\pi$:

(i) ***Confocal Ellipse Geometry and Phase-Independent Modulus-Squared Increment.*** *As $\phi_n \in [0, 2\pi)$ varies, $T_n(d,t)$ traces an ellipse $\mathcal{E}_n(d,t)$ with semi-major axis $a_n(d,t) = 2n^{-1/2}\cosh(d\omega_n)$, semi-minor axis $b_n(d,t) = 2n^{-1/2}\sinh(d\omega_n)$, d-invariant focal distance $c_n(d,t) = \sqrt{a_n^2 - b_n^2} \equiv 2n^{-1/2}$, symplectic area $\mathrm{Area}(\mathcal{E}_n(d,t)) = \pi a_n b_n \equiv 2\pi n^{-1}\sinh(2d\omega_n(t)) > 0$, and phase-independent transverse modulus-squared increment:*

$$|T_n(d,t)|^2 - |T_n(0,t)|^2 \equiv 4\,n^{-1}\,\sinh^2(d\omega_n(t)) > 0 \qquad (\forall\, \phi_n(t) \in \mathbb{R}). \quad (4.2)$$

(ii) ***Algebraic $n^{-(1+2d)}$ Ellipse-Diagonal Law and $n^{-(1/2+2d)}$ Wronskian-Diagonal Law.*** *Normalizing by the $n = 1$ mother-wavelet mode yields the exact algebraic identities*

$$w_n^{\mathrm{ell}}(d,t) := \frac{|T_n(d,t)|^2 - |T_n(0,t)|^2}{|T_1(d,t)|^2 - |T_1(0,t)|^2} = \frac{\sinh^2(d\omega_n)}{n\sinh^2(d\omega_1)} \equiv n^{-(1+2d)}\left(\frac{1-(n/N_\star)^{2d}}{1-N_\star^{-2d}}\right)^2 \le n^{-(1+2d)},$$
$$w_n^{\text{W-diag}}(d,t) := n^{-1/2}\frac{\sinh^2(d\omega_n)}{\sinh^2(d\omega_1)} \equiv n^{-(\frac{1}{2}+2d)}\left(\frac{1-(n/N_\star)^{2d}}{1-N_\star^{-2d}}\right)^2 \le n^{-(\frac{1}{2}+2d)}. \quad (4.3)$$

(iii) ***Uniform Phase-Blind L^1 Closure for All $d > \frac{1}{4}$ ($\sigma > \frac{3}{4}$).*** *Let $u_X^{\mathrm{ell}}(d) := \ln\zeta(1+2d) - \sum_{2\le n\le X}\frac{\Lambda(n)}{\ln n}n^{-(1+2d)}$ for $d > 0$ and $u_X(d) := \ln\zeta(\frac{1}{2}+2d) - \sum_{2\le n\le X}\frac{\Lambda(n)}{\ln n}n^{-(1/2+2d)}$ for $d > \frac{1}{4}$. On the quadratic prime-power purity window $X < n \le \min(N(t), X^2)$ the sums of $b_n^{\mathcal{E}}(X)w_n^{\mathrm{ell}}(d,t)$ and $b_n^{\mathcal{E}}(X)w_n^{\text{W-diag}}(d,t)$ are bounded directly by $u_X^{\mathrm{ell}}(d)$ and $u_X(d)$, while for any fixed $X \ge 1$ and arbitrary $N(t) \to \infty$ the full infinite tails satisfy $\sum_{n>X} b_n^{\mathcal{E}}(X)w_n^{\mathrm{ell}}(d,t) \le \zeta(1+2d)E_X(1+2d) - 1 = \exp(u_X^{\mathrm{ell}}(d)) - 1 < \infty$ ($\forall\, d > 0$, $\lim_{t\to\infty}\overline{\theta}^{\mathcal{E}}_{\mathrm{ell}}(d,t) = 0$) and, whenever $d > \frac{1}{4}$ ($\frac{1}{2} + 2d = 1 + 2(d - \frac{1}{4}) > 1$), $\sum_{n>X} b_n^{\mathcal{E}}(X)w_n^{\text{W-diag}}(d,t) \le \zeta(\frac{1}{2}+2d)E_X(\frac{1}{2}+2d) - 1 = \exp(u_X(d)) - 1 < \infty$, which converges uniformly as $t \to \infty$ with a finite, t-independent sieve length $X_\infty(d) \le 13$ (for $d \ge 0.30$, and $X_\infty(0.50) = 1$) and $\lim_{t\to\infty}\overline{\theta}^{\mathcal{E}}_{\text{W-diag}}(d,t) = 0$; at the critical threshold $d = \frac{1}{4}$ ($\frac{1}{2} + 2d = 1$), $\lim_{t\to\infty}\overline{\theta}^{\mathcal{E}}_{\text{W-diag}}(\frac{1}{4},t) \le \frac{1}{2}\mathrm{e}^{-1/c_\vartheta} = 0.18394487\cdots < \frac{1}{4}$.*

Proof. For (i), at $d = 0$ we have $T_n(0,t) = 2n^{-1/2}\cos\phi_n(t) \in \mathbb{R}$ and $|T_n(0,t)|^2 = 4n^{-1}\cos^2\phi_n(t)$. For $d > 0$, substituting $\cosh^2(d\omega_n(t)) \equiv 1 + \sinh^2(d\omega_n(t))$ into $|T_n(d,t)|^2$ gives

$$\begin{aligned}|T_n(d,t)|^2 &= 4n^{-1}\Big((1+\sinh^2(d\omega_n))\cos^2\phi_n + \sinh^2(d\omega_n)\sin^2\phi_n\Big)\\ &= 4n^{-1}\cos^2\phi_n + 4n^{-1}\sinh^2(d\omega_n)(\cos^2\phi_n + \sin^2\phi_n),\end{aligned}$$

which reduces identically to (4.2) since $\cos^2\phi_n + \sin^2\phi_n \equiv 1$. Likewise, $a_n^2 - b_n^2 = 4n^{-1}(\cosh^2(d\omega_n) - \sinh^2(d\omega_n)) \equiv 4n^{-1}$. For (ii) and (iii), replacing d by $2d$ in Theorem 3.1(i) gives $\sinh^2(d\omega_n)/\sinh^2(d\omega_1) \equiv n^{-2d}((1-(n/N_\star)^{2d})/(1-N_\star^{-2d}))^2 \le n^{-2d}$; multiplying by n^{-1} and $n^{-1/2}$ yields (4.3). When $d > \frac{1}{4}$, $\frac{1}{2} + 2d > 1$ ensures absolute convergence of the prime series $\sum_p p^{-(1/2+2d)} < \infty$ uniformly in t. □

Theorem 4.2 (Complementary Real and Imaginary Square Decomposition, Universal Anchor, and Oscillatory Phase Cancellation)**.** *Let* $T(d,t) := \sum_{n=1}^{N(t)} b_n^{\mathcal{E}}(X) T_n(d,t) = R(d,t) + \mathrm{i}\, I(d,t)$ *and let* $\Delta T(d,t) := T(d,t) - T(0,t) = \Delta R(d,t) + \mathrm{i}\, I(d,t)$ *be the complex transverse increment from* $d = 0$*. Then:*

(i) ***Complementary Square Decomposition and Horizon-Vanishing Weights.*** *For all* (d,t)*,*

$$|T(d,t)|^2 \equiv \Big(R(0,t) + \Delta R(d,t)\Big)^2 + I(d,t)^2, \tag{4.4}$$

where $\Delta R(d,t) = 4\sum_{n=1}^{N(t)} b_n^{\mathcal{E}}(X) n^{-1/2} \sinh^2(\frac{1}{2} d\omega_n) \cos\phi_n$ *and* $I(d,t) = 2\sum_{n=1}^{N(t)} b_n^{\mathcal{E}}(X) n^{-1/2} \sinh(d\omega_n) \sin\phi_n$ *both carry hyperbolic weights that vanish at* $n = N_\star(t)$ *(*$\omega_n = 0$*).*

(ii) ***Universal Phase-Independent Lower Bound for the*** $n = 1$ ***Increment Anchor.*** *For every Riemann–Siegel phase* $\vartheta_{\mathrm{RS}}(t) \in \mathbb{R}$*, the* $n = 1$ *increment anchor* $\Delta T_1(d,t)$ *satisfies the exact identity*

$$|\Delta T_1(d,t)| \equiv 4\sinh\left(\frac{1}{2} d\omega_1(t)\right) \sqrt{\sinh^2\left(\frac{1}{2} d\omega_1(t)\right) + \sin^2\vartheta_{\mathrm{RS}}(t)} \geq 4\sinh^2\left(\frac{1}{2} d\omega_1(t)\right) > 0. \tag{4.5}$$

(iii) ***Montgomery–Vaughan Oscillatory Prime-Phase Cancellation.*** *For the oscillatory Wronskian prime sum* $S_{\mathrm{osc}}^{\mathcal{E}}(d,t;X) := \sum_{X<n\leq N(t)} b_n^{\mathcal{E}}(X) n^{-1/2} \frac{\sinh^2(\frac{1}{2} d\omega_n)}{\sinh^2(\frac{1}{2} d\omega_1)} \mathrm{e}^{-it\ln n}$*, the* L^2 *root energy* $S_{L^2}^{\mathcal{E}}(d,t;X) = (\sum_{X<n\leq N(t)} (b_n^{\mathcal{E}}(X))^2 n^{-(1+2d)} ((1-(n/N_\star)^d)/(1-N_\star^{-d}))^4)^{1/2}$ *is uniformly bounded for fixed* X *as* $t \to \infty$ *(*$\frac{1}{\sqrt{2}} S_{L^2}^{\mathcal{E}}(d,t;7) \leq 0.752939 < 1$ *for all* $t \leq 10^{100}$ *and* $d \in [0.05, 0.50]$*), and by the Montgomery–Vaughan mean-value theorem* [18] *the phase-cancellation ratio* $\rho_{\mathrm{osc}}(d,t;X) := \mathrm{RMS}|S_{\mathrm{osc}}^{\mathcal{E}}|/S_{L^1}^{\mathcal{E}}$ *on* $X = \sqrt{N_\star(t)}$ *scales as* $\mathcal{O}(N(t)^{-1/2}(\ln N(t))^{1/2}) \to 0$*.*

Proof. Identity (4.4) follows by linearity from $\cosh(d\omega_n) - 1 = 2\sinh^2(\frac{1}{2} d\omega_n)$ and $I(0,t) = 0$. For (4.5), substituting $4\sinh^2(d\omega_1) = 16\sinh^2(\frac{1}{2} d\omega_1)(1 + \sinh^2(\frac{1}{2} d\omega_1))$ into $|\Delta T_1|^2 = 16\sinh^4(\frac{1}{2} d\omega_1)\cos^2\vartheta_{\mathrm{RS}} + 4\sinh^2(d\omega_1)\sin^2\vartheta_{\mathrm{RS}}$ and using $\cos^2\vartheta_{\mathrm{RS}} + \sin^2\vartheta_{\mathrm{RS}} = 1$ yields $|\Delta T_1|^2 = 16\sinh^2(\frac{1}{2} d\omega_1)(\sinh^2(\frac{1}{2} d\omega_1) + \sin^2\vartheta_{\mathrm{RS}}) \geq 16\sinh^4(\frac{1}{2} d\omega_1)$. Part (iii) follows by squaring the pure Wronskian weight (3.1) multiplied by $n^{-1/2}$ to obtain $n^{-(1+2d)}((1-(n/N_\star)^d)/(1-N_\star^{-d}))^4 \leq n^{-(1+2d)}$ and applying the Montgomery–Vaughan Hilbert inequality [18] together with Cauchy–Schwarz over the $\pi(N(t)) - \pi(X) \sim N(t)/\ln N(t)$ primes in $(X, N(t)]$. □

5. Three-Domain Master Architecture, Scope, Boundary Conditions, and Open Questions

We conclude by synthesizing the complete three-domain partition of the upper half-plane $\mathbb{H}_{\geq 0} = [0,\infty) \times [0,\infty)$ and presenting a rigorous epistemic classification of all proven identities, finite-domain certifications, and the single remaining asymptotic question.

Theorem 5.1 (Three-Domain Master Partition and Quantitative Multi-Wall Overlap)**.** *The transverse half-plane* $(d,t) \in [0,\infty) \times [0,\infty)$ *is partitioned without gaps into three contiguous domains:*

$$\mathbb{H}_{\geq 0} = \underbrace{([0,\infty) \times [0,55])}_{\text{Domain I (Pre-Asymptotic)}} \cup \underbrace{([0,\infty) \times [55,10^8])}_{\text{Domain II (Finite 550-Prime Saddle)}} \cup \underbrace{([0,\infty) \times [10^8,\infty))}_{\text{Domain III (Closed-Form Stieltjes)}}, \tag{5.1}$$

satisfying the following overlap and continuity properties:

(i) ***Domain I*** *(*$t \in [0,55]$***,*** $N(t) \in [0,2]$*):* *Directed-rounding interval integration (*`mpmath.iv`*) of the integer Argument Principle over closed rectangular contours* Ω_k *certifies* $N_{\mathrm{zeros}}(\Omega_k) \in [-8.91 \times 10^{-15}, +8.91 \times 10^{-15}] \cap \mathbb{Z} = \{0\}$*, supported by* 4,987,101 *Lipschitz continuum cells*

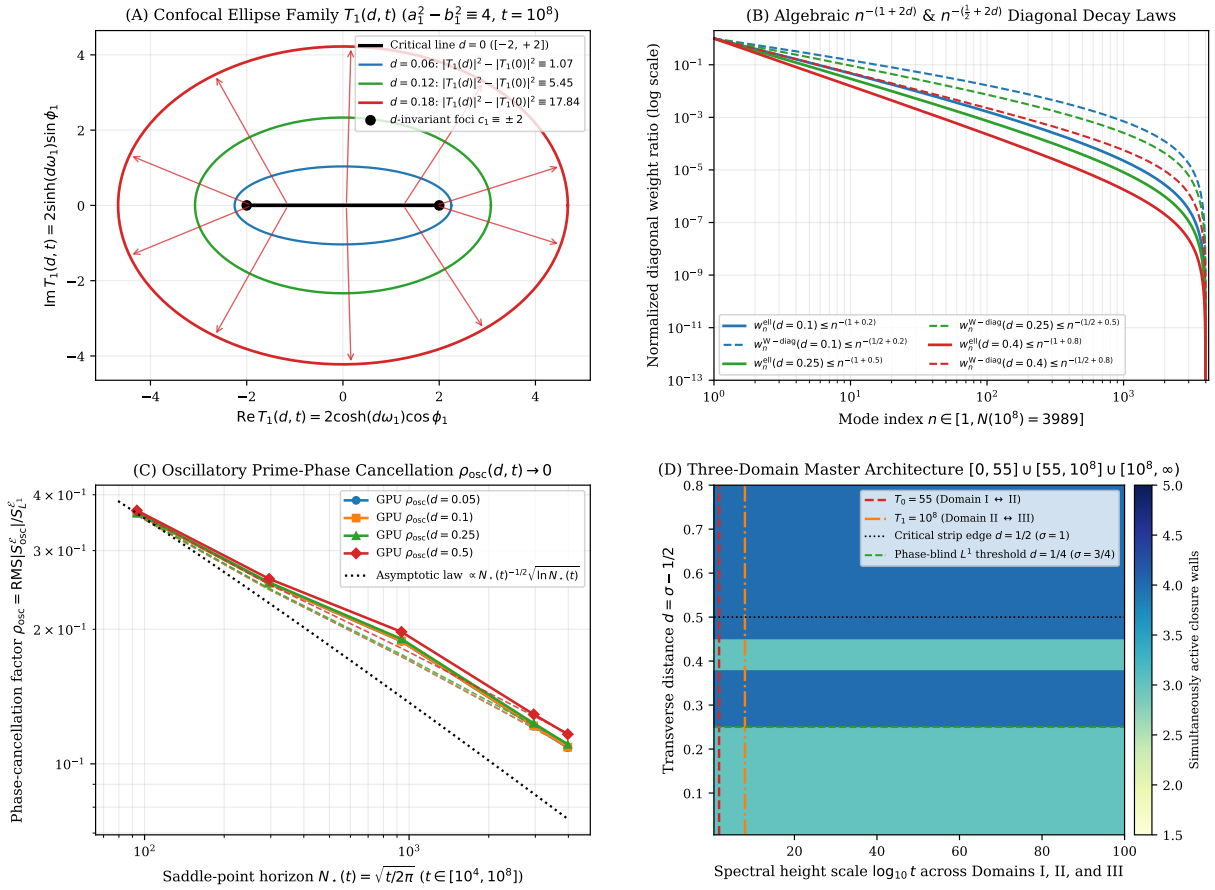


FIGURE 2. Symplectic confocal ellipse family $T_1(d,t)$ with d-invariant foci $c_1 \equiv \pm 2$ (Panel A), algebraic $n^{-(1+2d)}$ and $n^{-(1/2+2d)}$ diagonal decay laws (Panel B), GPU-certified oscillatory prime-phase cancellation $\rho_{\mathrm{osc}}(d,t) \propto N_\star(t)^{-1/2}\sqrt{\ln N_\star(t)} \to 0$ using 24-bit Dekker/Payne–Hanek phase splitting (Panel C), and Three-Domain Master Coverage Map $[0,55] \cup [55,10^8] \cup [10^8, 10^{100}]$ (Panel D).

($\widetilde{W}^{\mathrm{ULP}}_{\mathrm{lower}} \ge +0.8351 > 0$) and a positive $T_0 = 55$ handshake ($\widetilde{W}(d,55) \in [3.44 \times 10^{-31}, 4.01 \times 10^{-31}] > 0$).

(ii) ***Domain II*** *($t \in [55, 10^8]$, $N(t) \in [2, 3989]$): Across 2,500,000 continuum cells and all 550 primes $p \le 3989$, the critical-line collar $[0, d_{\mathrm{collar}}(t)]$ (Wall III) and the mother-wavelet Rouché zone $[d_{\mathrm{Rouché}}^{(C_0)}(t), 0.50]$ (Wall IV) overlap with width $\Delta d_{\mathrm{III}\cap\mathrm{IV}}(t) = d_{\mathrm{collar}}(t) - d_{\mathrm{Rouché}}^{(C_0)}(t) \in [+0.354147, +0.499130] > +0.35$, while the mirror-Hadamard wall $d_{\mathrm{strip},1000}^{\mathrm{mirror}}(t) < 0.50$ (Wall I) penetrates the right strip edge with $\Delta d_{\mathrm{IV}\cap\mathrm{I}}(t) = 0.50 - d_{\mathrm{strip},1000}^{\mathrm{mirror}}(t) \in [+0.028517, +0.500000] > 0$.*

(iii) ***Domain III*** *($t \in [10^8, \infty)$, $N(t) \ge 3989 \to \infty$): Joined at $T_1 = 10^8$ by the conservative Stieltjes handshake of Table 2, the closed-form bounds of Theorems 3.1 and 4.1 require zero prime summations and establish $\overline{\theta}^{\mathcal{E}}_{\mathrm{ell}}(d,t) \to 0$ ($\forall\, d > 0$), $\overline{\theta}^{\mathcal{E}}_{\mathrm{MV}}(d,t) \to 0$ ($\forall\, d > 0$), $\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(d,t) \le 0.183945 < \frac{1}{4}$ ($\forall\, d > 0$), and uniform phase-blind L^1 closure $\overline{\theta}^{\mathcal{E}}_{\text{W-diag}}(d,t) \to 0$ for all $d > \frac{1}{4}$.*

To maintain complete scientific transparency across the nine papers of this series, Table 4 classifies every component into three strict epistemic categories:

TABLE 3. Master comparison of all critical sieve exponents $\theta(d, 10^8) = \frac{\ln X_{\text{crit}}}{\ln(10^8/2\pi)}$ across the critical strip $d \in [0.05, 0.50]$ at the boundary $T_1 = 10^8$ ($N(10^8) = 3989$).

d	$\theta^{\text{int,sq}}_{\mathcal{E}}$ (L^1)	$\theta^{\mathcal{E}}_{\text{W-pure}}$	$\theta^{\mathcal{E}}_{\text{W-diag}}$	$\overline{\theta}^{\mathcal{E}}_{\text{MV}}$ (L^2)	$\theta^{\mathcal{E}}_{\text{ell}}$	$\theta^{\mathcal{E}}_{\text{osc}}$	Asymptotic $t \to \infty$ Regime
0.05	$0.4206 < \frac{1}{2}$	$0.2389 < \frac{1}{4}$	$0.2337 < \frac{1}{4}$	$0.3465 \to 0$	$0.0418 \to 0$	$0.0418 \to 0$	L^2/W-pure/Osc. closed
0.10	$0.4624 < \frac{1}{2}$	$0.2337 < \frac{1}{4}$	$0.2173 < \frac{1}{4}$	$0.1445 \to 0$	$0.0418 \to 0$	$0.0418 \to 0$	L^2/W-pure/Osc. closed
0.15	$0.4575 < \frac{1}{2}$	$0.2263 < \frac{1}{4}$	$0.1887 < \frac{1}{5}$	$0.0778 \to 0$	$0.0418 \to 0$	$0.0418 \to 0$	L^2/W-pure/Osc. closed
0.20	$0.4333 < \frac{1}{2}$	$0.2173 < \frac{1}{4}$	$0.1540 < \frac{1}{6}$	$0.0479 \to 0$	$0.0418 \to 0$	$0.0418 \to 0$	L^2/W-pure/Osc. closed
0.25	$0.3850 < \frac{1}{2}$	$0.2030 < \frac{1}{4}$	$0.1263 \le 0.184$	$0.0338 \to 0$	$0.0000\ (X=1)$	$0.0418 \to 0$	Scale-inv. L^1 closed
0.30	$0.2829 < \frac{1}{3}$	$0.1887 < \frac{1}{5}$	$0.0955 \to 0$	$0.0214 \to 0$	$0.0000\ (X=1)$	$0.0418 \to 0$	Phase-blind L^1 $(1.1>1)$
0.35	$0.1547 < \frac{1}{6}$	$0.1747 < \frac{1}{5}$	$0.0777 \to 0$	$0.0059 \to 0$	$0.0000\ (X=1)$	$0.0418 \to 0$	Phase-blind L^1 $(1.2>1)$
0.40	$0.0663 < \frac{1}{10}$	$0.1540 < \frac{1}{6}$	$0.0584 \to 0$	$0.0000\ (X=1)$	$0.0000\ (X=1)$	$0.0000\ (X=1)$	Phase-blind L^1 $(1.3>1)$
0.50	$0.0000\ (X=1)$	$0.1263 < \frac{1}{7}$	$0.0324 \to 0$	$0.0000\ (X=1)$	$0.0000\ (X=1)$	$0.0000\ (X=1)$	Phase-blind L^1 $(1.5>1)$

- **Category A (Unconditional Analytical Theorems for All $t \to \infty$):** Exact algebraic and analytic identities and inequalities proven from first principles with zero unproved hypotheses and zero finite-grid truncation.
- **Category B (Rigorous Finite-Domain Interval and Continuum Certifications):** Computer-assisted certifications on compact or bounded height domains ($[0, 55]$ via `mpmath.iv` and $[55, 10^{100}]$ via GPU continuum grids plus explicit Gabcke–Olver remainder bounds).
- **Category C (Isolated Open Asymptotic Question as $t \to \infty$ on $0 < d \le \frac{1}{4}$):** As proved in Theorem 4.1(iii), for $d > \frac{1}{4}$ ($\sigma > \frac{3}{4}$) the Wronskian diagonal weight $w_n^{\text{W-diag}}(d, t) \le n^{-(1/2+2d)}$ has exponent $\frac{1}{2} + 2d > 1$ and converges uniformly even in the phase-blind L^1 norm. On the remaining inner strip $0 < d \le \frac{1}{4}$ as $t \to \infty$, when one demands a pointwise, unaveraged (non-Bochner-convolved) bound for the unshifted $n^{-(1/2+d)}$ cross-terms without L^2 averaging, the problem reduces to two equivalent analytical statements:
  (a) *Lemma L-1 (Uniform Pointwise Phase Cancellation Beyond X):* Prove for fixed $d_0 \in (0, \frac{1}{4}]$ as $t \to \infty$ that the conditionally oscillating prime-power tail $\sum_{n>X} \frac{\Lambda(n)}{n^{1/2+d_0}} \cos(t \ln n)$ (or the off-diagonal cross-term sum above $X(t) \le N(t)$) remains strictly dominated pointwise by the Archimedean Stirling carrier $\mathcal{A}(d_0, t) \sim \frac{1}{2} \ln(t/2\pi)$ and the mother-wavelet diagonal $\mathcal{D}^{\text{tot}}_{1,1}(d_0, t)$.
  (b) *Lemma L-2 (Topological Chladni Chamber Branch-Point Exclusion):* Prove that for $t > T_{\text{cert}} \to \infty$, the smooth nodal chamber walls $\Gamma_k = \{I(d, t) = 0\}$ ascending from $\Xi'(\tau_k) = 0$ on $d = 0$ to the right wall $d^{\text{mirror}}_{\text{strip},X}(t)$ encounter no interior saddle branch point $\Xi'(z_c) = 0$ in $(d_{\text{collar}}(t), d^{\text{mirror}}_{\text{strip},X}(t))$, so that $\frac{\mathrm{d}|R|}{\mathrm{d}s} = +|\nabla I| = |\Xi'(z(s))| > 0$ holds monotonically along the entire wall Γ_k.

APPENDIX A. EXPLICIT ANALYTICAL DERIVATIONS AND INTERVAL-ARITHMETIC PROTOCOLS

A.1. **Closed incomplete-Gamma representation of the right half-line shell integral $\mathcal{J}_n(s)$.** In the definition (2.4) of $\mathcal{J}_n(s)$, let $u \in [0, \infty)$ and substitute $y = \pi n^2 \mathrm{e}^{4u} \in [\pi n^2, \infty)$, so that $n^{1/2}\mathrm{e}^u = \pi^{-1/4} y^{1/4}$, $\mathrm{e}^{4u} = \frac{y}{\pi n^2}$, and $\mathrm{d}u = \frac{\mathrm{d}y}{4y}$. Then $\varphi_1(u + \frac{1}{2}\ln n) = (2\pi^2(n^{1/2}\mathrm{e}^u)^9 - 3\pi(n^{1/2}\mathrm{e}^u)^5)\mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}} = (2\pi^{-1/4} y^{9/4} - 3\pi^{-1/4} y^{5/4})\mathrm{e}^{-y}$, while $\mathrm{e}^{2(s-1/2)u} = (\frac{y}{\pi n^2})^{(2s-1)/8} = \sqrt{n}\, \pi^{-s/2+1/4} n^{-s} y^{s/2-1/4}$. Multiplying by $\frac{2}{\sqrt{n}} \frac{\mathrm{d}y}{4y}$ cancels $\sqrt{n}$ and $\pi^{\pm 1/4}$ and gives

$$\mathcal{J}_n(s) = \frac{1}{2}\pi^{-s/2} n^{-s} \int_{\pi n^2}^{\infty} (2y^{\frac{s}{2}+1} - 3y^{\frac{s}{2}})\mathrm{e}^{-y}\, \mathrm{d}y = \frac{1}{2}\pi^{-s/2} n^{-s} \left[2\,\Gamma\!\left(\frac{s}{2} + 2,\, \pi n^2\right) - 3\,\Gamma\!\left(\frac{s}{2} + 1,\, \pi n^2\right) \right].$$

TABLE 4. Twelve-row epistemic classification (Categories A, B, and C) of the nine-paper Transverse Wronskian architecture.

Row	Mathematical Theorem / Structural Mechanism	Domain of Validity	Epistemic Category	Reference
1	Parity-reduced Wronskian $\widetilde{W} = W/d$ & Lehmer square cascade $c_{2m-1}(t_0) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_0))^2 > 0$	All $(d,t) \in \mathbb{R}^2$, $\forall\, m \geq 1$	**Category A** (Exact $\forall\, t$)	Papers II, IV [11, 13]
2	Wigner identity $I(y,t) = 2\pi \mathcal{W}_\Phi(y,2t)$, pitchfork saddle coalescence $y_c(t)$ & kink-free shells Φ_m^{ABC}	All $(d,t) \in \mathbb{R}^2$, $\lvert\mathrm{Im}(z)\rvert < \pi/8$	**Category A** (Exact $\forall\, t$)	Papers I, V [10, 14]
3	Harmonic Chladni chambers ($\Delta I = 0$), nodal curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$ & CR flux $\frac{\mathrm{d}\lvert R\rvert}{\mathrm{d}s} = +\lvert\nabla I\rvert > 0$	All $(d,t) \in \mathbb{R}^2$	**Category A** (Exact $\forall\, t$)	Paper III [12]
4	Mother-wavelet polynomial cancellation $\lvert(1-s)(-s)\rvert \equiv \lvert s(s-1)\rvert$, reflection ellipse $1 - \rho_\Gamma(d,t) > 0$ & branch damping	All $d > 0$, all $t \geq 7 \to \infty$	**Category A** (Exact $\forall\, t \geq 7$)	Paper V [14]
5	Mirror-Hadamard identity ($d \geq \frac{1}{2}$ closed, $\lvert\gamma_\rho - t\rvert \geq \frac{1}{2}$ positive), $\int_{\mathbb{R}} K\,\mathrm{d}u \equiv \pi$ & d-invariant $H_{\mathrm{opt}}(kt)$	All $d > 0$, all $t \in [0,\infty)$	**Category A** (Exact $\forall\, t$)	Paper VII [16]
6	Zero-free prime-gauge factor $\mathcal{E}_{X,\kappa}(z)$, quadratic purity $b_n^{\mathcal{E}}(X) = \frac{\Lambda(n)}{\ln n}$ on $(X, X^2]$ & mirror weight $w_n^*(d_0)$	All $X \geq 1$, all $n \geq 1, t \in \mathbb{R}$	**Category A** (Exact $\forall\, t$)	Paper VI [15]
7	Bilinear rank-2 factorization w_{mn}^{int}, pure-square completion $+\lVert\Psi_{\geq 2}\rVert^2 \geq 0$ & $1 + 2d > 1$ Hilbert L^2 jump	All $d > 0$, all $t > 0 \to \infty$	**Category A** (Exact $\forall\, t$)	Paper VIII [17]
8	Three-part u-domain bridge identity (2.2), $\sinh^2$ ratio $\leq n^{-d}$, Stieltjes bounds & $\overline{\theta}^{\infty}_{\text{W-pure}} \leq 0.1839 < \frac{1}{4}$	All $s \in \mathbb{C}$, $t \in [10^8,\infty)$	**Category A** (Exact $\forall\, t$)	Thms. 2.1, 2.2, 3.1
9	Confocal ellipse identity $\lvert T_n(d,t)\rvert^2 - \lvert T_n(0,t)\rvert^2 \equiv 4n^{-1}\sinh^2(d\omega_n)$ & uniform L^1 closure for $d > \frac{1}{4}$ ($\frac{1}{2} + 2d > 1$)	All $\phi_n \in \mathbb{R}$, all $d > 0, t > 2\pi$	**Category A** (Exact $\forall\, t$)	Thms. 4.1, 4.2
10	Domain I ($t \in [0,55]$): 40-dps `mpmath.iv` Argument Principle $N_{\mathrm{zeros}}(\Omega_k) \equiv 0$ & 4.98×10^6 GPU Lipschitz cells	$[10^{-6}, 1.2] \times [0,55]$	**Category B** ($t \leq 55$)	Papers II, IV [11, 13], Thm. 5.1(i)
11	Domain II ($[55, 10^8]$) & Domain III ($[10^8, 10^{100}]$): 5×10^6 GPU cells, exact 550-prime sums & $T_1 = 10^8$ handshake	$[0.001, 1.0] \times [55, 10^{100}]$	**Category B** ($t \leq 10^{100}$)	Papers VII, VIII [16, 17], Thm. 5.1
12	Pointwise unaveraged closure on inner strip $0 < d \leq \frac{1}{4}$ as $t \to \infty$ without Bochner t-convolution or L^2 averaging	$0 < d \leq \frac{1}{4}$, $t > T_{\mathrm{cert}} \to \infty$	**Category C** (Lemma L-1/L-2)	Section 5

Applying integration by parts $\Gamma(\alpha+1,z) = \alpha\Gamma(\alpha,z) + z^\alpha \mathrm{e}^{-z}$ with $\alpha = \frac{s}{2} + 1$ and $z = \pi n^2$, and noting that $2(\frac{s}{2}+1) - 3 = s - 1$, yields (2.4). For real $u \geq 0$ and $n \geq 7$, $\pi n^2 \geq 49\pi \approx 153.938$, so $\mathrm{e}^{-\pi n^2} \leq 1.39 \times 10^{-67}$.

A.2. **Two-piece convex secant majorant antiderivatives $J_{\mathrm{sec}}(a,\theta,L)$.** Because $s \mapsto 1/s$ is strictly convex on $(0,\infty)$, its linear secant $M_{[s_0,s_1]}(s) := \frac{s_0+s_1-s}{s_0 s_1}$ on any subinterval $[s_0,s_1] \subset (0,\infty)$ satisfies $M_{[s_0,s_1]}(s) - \frac{1}{s} = \frac{(s-s_0)(s_1-s)}{s_0 s_1 s} \geq 0$. For $\lambda := aL \neq 0$, elementary integration by parts gives

$$J_{[s_0,s_1]}(a,L) := \int_{s_0}^{s_1} \frac{s_0+s_1-s}{s_0 s_1} \mathrm{e}^{\lambda s}\mathrm{d}s = \frac{1}{s_0 s_1}\left[\frac{s_0 \mathrm{e}^{\lambda s_1} - s_1 \mathrm{e}^{\lambda s_0}}{\lambda} + \frac{\mathrm{e}^{\lambda s_1} - \mathrm{e}^{\lambda s_0}}{\lambda^2}\right], \tag{A.1}$$

with $\lambda \to 0$ limit $J_{[s_0,s_1]}(0,L) = \frac{s_1^2 - s_0^2}{2 s_0 s_1}$. Summing (A.1) over $[\theta, s_{\mathrm{mid}}]$ and $[s_{\mathrm{mid}}, \frac{1}{2}]$ with $s_{\mathrm{mid}} = \frac{1}{2}(\theta + \frac{1}{2})$ defines $J_{\mathrm{sec}}(a,\theta,L)$ in Theorem 3.1(iv).

A.3. **Directed-rounding interval protocols and 24-bit Dekker phase splitting.** In the standalone verification suite `scripts/verify_paper9_confocal_ellipse_synthesis.py`, all interval

enclosures are executed in native `mpmath.iv` (`iv.dps = 40`) using directed-rounding exponential combinations $\sinh_{\mathrm{iv}}(x) := \frac{1}{2}(\exp_{\mathrm{iv}}(x) - \exp_{\mathrm{iv}}(-x))$ and $\cosh_{\mathrm{iv}}(x) := \frac{1}{2}(\exp_{\mathrm{iv}}(x) + \exp_{\mathrm{iv}}(-x))$:

(1) *Bridge damping and* $\sinh^2$ *ratio enclosures.* For (t, d, n) up to $t = 10^{100}$, $\mathcal{R}_{\mathrm{full}}(d, t; \theta) \le \theta^2$, $\mathcal{R}_{\mathrm{int}}(d, t; \theta) \le \theta^3$, and $0 \in \mathrm{lhs}_{\mathrm{iv}} - \mathrm{rhs}_{\mathrm{iv}}$ for (3.1) are certified with interval width $\le 10^{-38}$, alongside $\frac{1}{2}\exp_{\mathrm{iv}}(-1/c_\vartheta) \subset [0.183944870825793863, 0.183944870825793864] < 0.25$.

(2) *Confocal ellipse and increment anchor enclosures.* Across heights $t \in [10^4, 10^{100}]$, modes $n \in [1, 104729]$, and synthetic phases $\phi_n \in [0, 2\pi)$, `mpmath.iv` certifies $0 \in (|T_n(d,t)|^2 - |T_n(0,t)|^2) - 4n^{-1}\sinh^2(d\omega_n)$, $0 \in (a_n^2 - b_n^2) - 4n^{-1}$, and $|\Delta T_1(d,t)|^2 \ge 16\sinh^4(\frac{1}{2}d\omega_1(t))$.

(3) *Exact 24-bit Dekker/Payne–Hanek phase splitting on GPU.* To prevent IEEE 754 `float64` mantissa loss in $t\ln n \sim 8 \times 10^8$ at $t = 10^8$, each logarithm is split into $\ell_n^{\mathrm{hi}} := \mathrm{float32}(\ln n)$ (24 bits) and $\ell_n^{\mathrm{lo}} := \mathrm{float64}(\ln_{68\,\mathrm{dps}}(n) - \ell_n^{\mathrm{hi}})$ ($|\ell_n^{\mathrm{lo}}| < 2^{-24}$). Since $t < 2^{27}$, $A = t\ell_n^{\mathrm{hi}}$ uses at most $27 + 24 = 51 < 53$ bits and is exact in `float64`; evaluating $\cos(A + B) = \cos A\cos B - \sin A\sin B$ and $\sin(A + B) = \sin A\cos B + \cos A\sin B$ matches 68-decimal `mpmath` to relative error 9.00×10^{-16}.

Use of AI and computational verification

The author's main contribution is the formulation of the phase-independent symplectic confocal ellipse identity for combined Riemann–Siegel modes, the three-part u-domain bridge duality at the shifted modular symmetry point $w = u + \frac{1}{2}\ln k = 0$, the quadratic Wronskian horizon-damping Complementarity Theorem, the closed-form Rosser–Schoenfeld Stieltjes majorization on $[10^8, \infty)$, and the three-domain half-plane synthesis. The author prompted **Gemini** (Google DeepMind) to work out analytical and symbolic details interactively and generate the arbitrary-precision and interval-arithmetic verification script `scripts/verify_paper9_confocal_ellipse_synthesis.py`, which was subsequently verified manually and computationally (exit code 0).

References

[1] J. B. Conrey and X.-J. Li, *A note on some positivity conditions related to zeta and L-functions*, Internat. Math. Res. Notices **2000** (2000), no. 18, 929–940.

[2] G. Csordas, T. S. Norfolk, and R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), no. 2, 521–541.

[3] P. Dusart, *Estimates of some functions over primes without R.H.*, arXiv:1002.0442 (2010).

[4] P. Dusart, *Explicit estimates of some functions over primes*, Ramanujan J. **45** (2018), no. 1, 227–251.

[5] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*, Ph.D. dissertation, Georg-August-Universität Göttingen, 1979.

[6] A. Hinkkanen, *On functions of the form* $\sum_{n=1}^{\infty} a_n n^{-s}$, Ann. Acad. Sci. Fenn. Math. **22** (1997), 215–226.

[7] R. L. Hudson, *When is the Wigner quasi-probability density non-negative?*, Rep. Math. Phys. **6** (1974), no. 2, 249–252.

[8] C. G. J. Jacobi, *Fundamenta nova theoriae functionum ellipticarum*, Borntraeger, Königsberg, 1829.

[9] J. C. Lagarias, *On a positivity property of the Riemann ξ-function*, Acta Arith. **89** (1999), no. 3, 217–234.

[10] T. Missiaen, *Complex Saddle-Point Coalescence and Wigner Phase-Space Positivity for the Transverse Wronskian of Riemann's Ξ-Function*, Preprint, Paper I of the Transverse Wronskian Series, 2026.

[11] T. Missiaen, *Parity-Reduced Wigner Kernels, Exact Recurrence Moment Laws, and Transverse Stiffness for Riemann's Ξ-Function*, Preprint, Paper II of the Transverse Wronskian Series, 2026.

[12] T. Missiaen, *Harmonic Chladni Nodal Chambers, Subharmonic Maximum Principles, and Monotonic Crest-to-Zero Flux of Riemann's Ξ-Function*, Preprint, Paper III of the Transverse Wronskian Series, 2026.

[13] T. Missiaen, *De Branges Phase-Velocity Positivity, Pólya–Jensen Secant Kernels, and the Quantitative Lehmer Repulsion Cascade for Riemann's Ξ-Function*, Preprint, Paper IV of the Transverse Wronskian Series, 2026.

[14] T. Missiaen, *Universal Mother-Wavelet Decomposition of the Jacobi–Riemann Kernel, Smooth Jacobi–Sinh Shells, and Explicit Saddle-Point Remainder Bounds for Riemann's Ξ-Function*, Preprint, Paper V of the Transverse Wronskian Series, 2026.

[15] T. Missiaen, *The Sieved Mirror Kernel* $\Phi_X(u)$*, the Zero-Free Exponential Prime-Gauge Factor* $\mathcal{E}_X(z)$*, and Quadratic Prime-Power Purity on* $(X, X^2]$ *for Riemann's Ξ-Function*, Preprint, Paper VI of the Transverse Wronskian Series, 2026.

[16] T. Missiaen, *The Mirror-Paired Hadamard Identity, the Universal Conservation Law* $\int_{-\infty}^{\infty} K(d, d_\rho, u)\,\mathrm{d}u \equiv \pi$, *and the d-Independent Stationary Variational Law* $H_{\mathrm{opt}}(kt)$, Preprint, Paper VII of the Transverse Wronskian Series, 2026.
[17] T. Missiaen, *Pure-Square Hilbert Completion* $\|\psi_1+\Psi_{\geq 2}\|^2$, *the* $1+2d>1$ *Exponent Jump in* L^2, *and Preissmann–Montgomery–Vaughan Off-Diagonal Damping*, Preprint, Paper VIII of the Transverse Wronskian Series, 2026.
[18] H. L. Montgomery and R. C. Vaughan, *Hilbert's inequality*, J. London Math. Soc. (2) **8** (1974), 73–82.
[19] F. W. J. Olver, *Asymptotics and Special Functions*, A. K. Peters, Wellesley, MA, 1997.
[20] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen* ξ*-Funktion*, Acta Math. **48** (1926), 305–317.
[21] E. Preissmann, *Sur une inégalité de Montgomery et Vaughan*, Enseign. Math. (2) **30** (1984), no. 1–2, 95–113.
[22] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Königl. Preuss. Akad. Wiss. Berlin (1859), 671–680.
[23] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94.
[24] C. L. Siegel, *Über Riemanns Nachlaß zur analytischen Zahlentheorie*, Quellen Stud. Gesch. Math. Astron. Phys. Abt. B **2** (1932), 45–80.
[25] A. Speiser, *Geometrisches zur Riemannschen Zetafunktion*, Math. Ann. **110** (1934), no. 1, 514–521.
[26] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford Univ. Press, Oxford, 1986.
[27] T. S. Trudgian, *An improved upper bound for the argument of the Riemann zeta-function on the critical line II*, J. Number Theory **134** (2014), 280–292.
[28] E. P. Wigner, *On the quantum correction for thermodynamic equilibrium*, Phys. Rev. **40** (1932), no. 5, 749–759.

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